WJEC

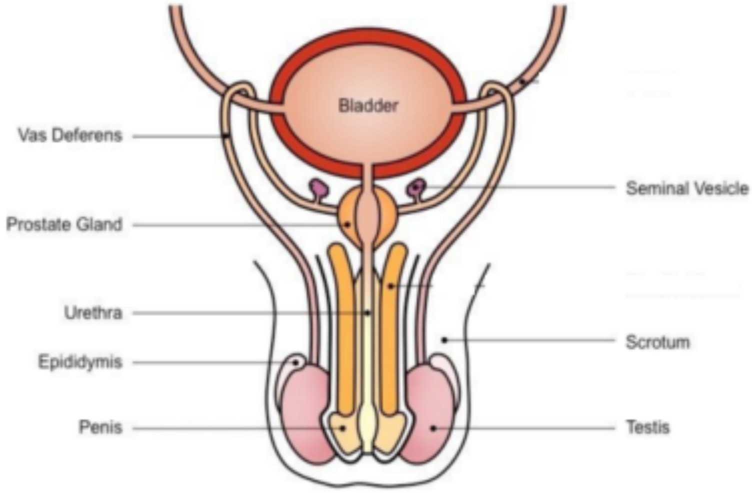
Biology A2

Unit 4

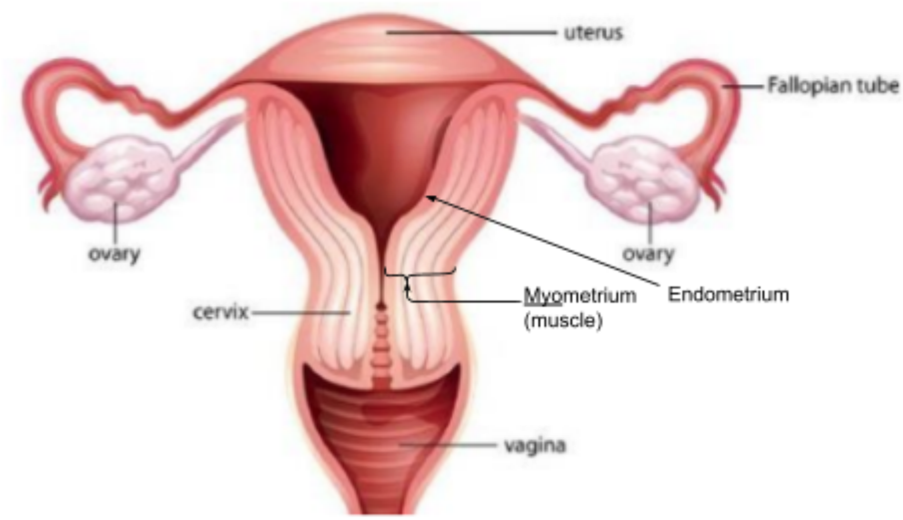
Version 1.5: 4.1, 4.2 (intro), 4.3, 4.4, 4.5 (intro)
Content are for mock 3

Ian Lam

4.1 Reproduction in Humans

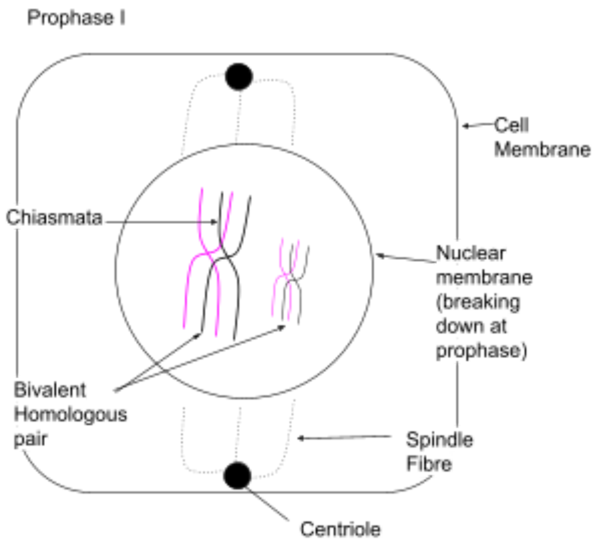


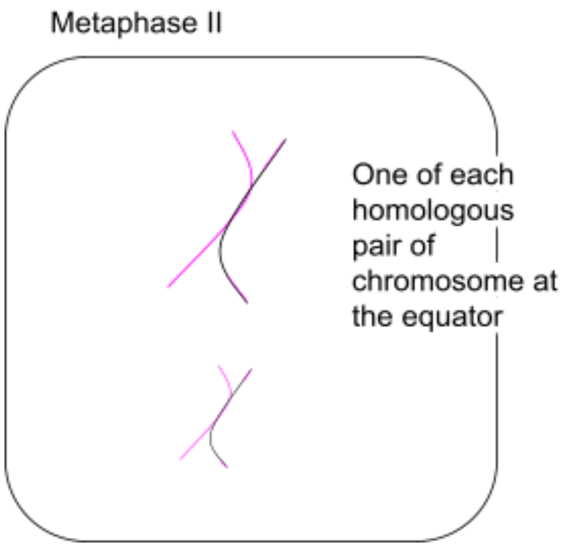
Structure Function	**Structure Function**
Scrotum	• Testes outside of the body (sperm production better below body temp)
Testes	• Seminiferous tubule the male gametes (spermatozoa) are made • Produce the hormone testosterone (male sex hormone) in leydig cell
Epididymis	• Sperm mature here before entering the vas deferens. • Storage of sperm
Vas Deferens	• Carries sperm to the urethra by peristalsis
Seminal Vesicles	• Secrete mucus and a watery alkaline fluid containing nutrients (Including fructose – energy source for respiration / ATP production) • Empties its contents into an ejaculatory duct (during ejaculation)
Prostate gland	• Secretes mucus and an alkaline fluid into the ejaculatory duct. • Helps to neutralise the acidity of the vagina (reduce bacteria growth)
Urethra	• Carries urine from the bladder and semen out of the penis.
Penis	• Contains erectile tissue, which fills with blood causing the Penis to become erect.



Structure Function	Structure Function
Ovaries	● Where the ova are made ● Secrete oestrogen and progesterone (female sex hormone)
Fallopian tubes (Oviducts)	● Carry ova from the ovaries to uterus – lined with ciliated epithelium ● Site of fertilisation
Uterus	● Foetus develops during pregnancy ● Has muscular walls which contract during labour.
Endometrium (Uterus Lining)  	● The inner most layer of the uterus ● The site of embryo implantation ● Contains blood vessels and glands ● Gets shed during menstruation. ● Site of exchange
Cervix ‘neck’	● The narrow entrance to the uterus from the vagina. ● A ring of muscle which can open/close ● Often blocked by a plug of mucus
Vagina	● A muscular tube containing elastic tissue ● Stretches during childbirth and sexual intercourse (birth canal)

Recap on Cell Division





<u>Sperm</u>

Interphase → Telophase

<u>Eggs</u>

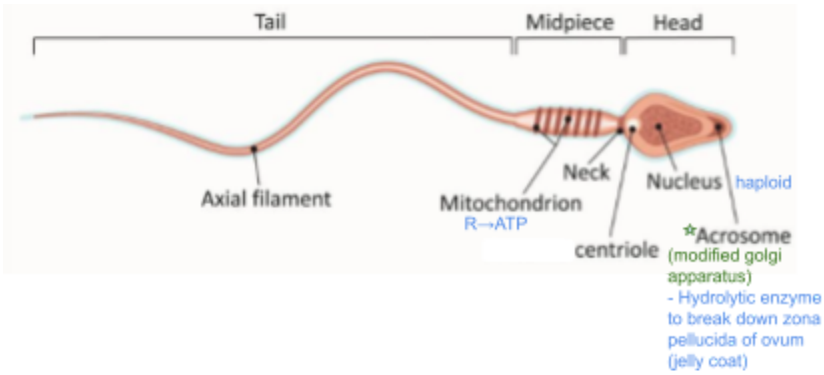
Interphase → Prophase I → Metaphase II → Telophase

Before birth   Ovulation

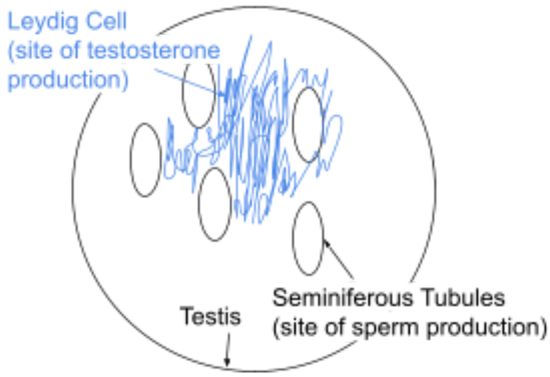
Gametogenesis

Gametogenesis is the formation of haploid gametes.

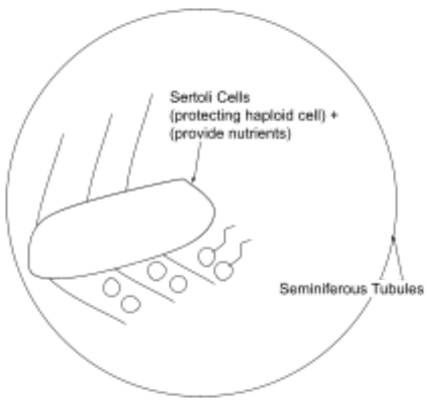
Spermatozoa



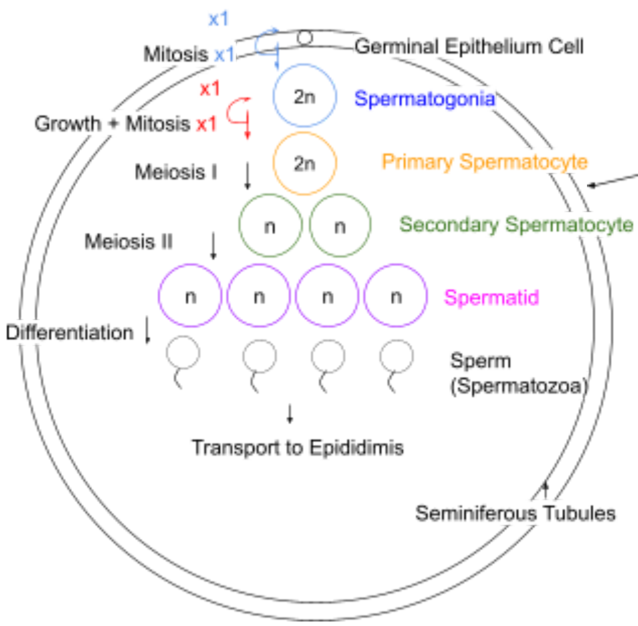
Testis



Seminiferous Tubules

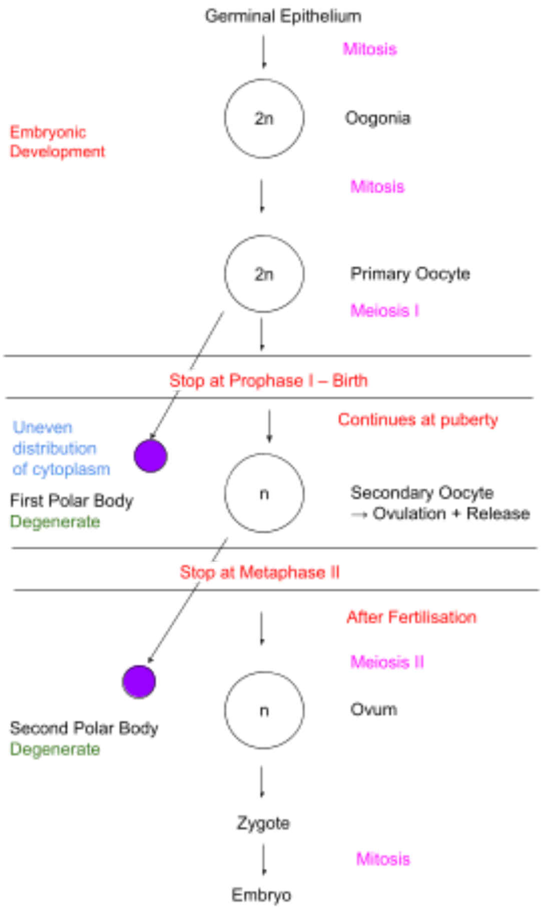


Spermatogenesis – Males

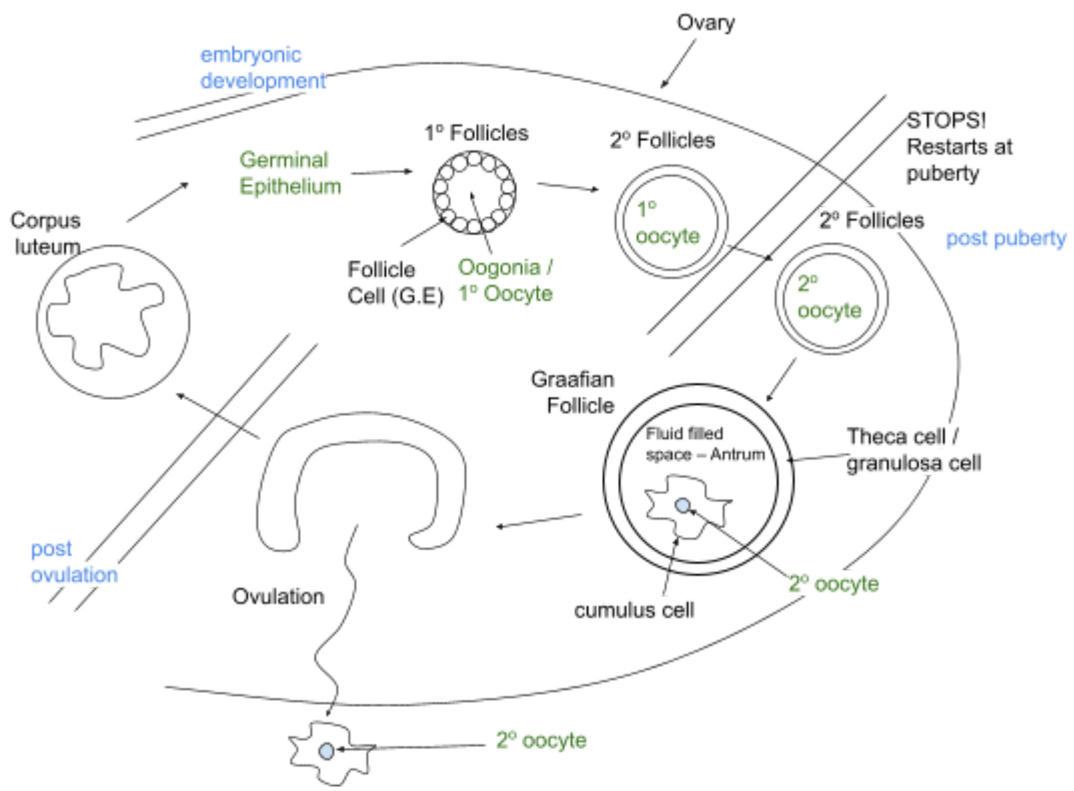


<u>Mitosis</u> helps maintain cell membranes so these tissue don't run out

Oogenesis – Females

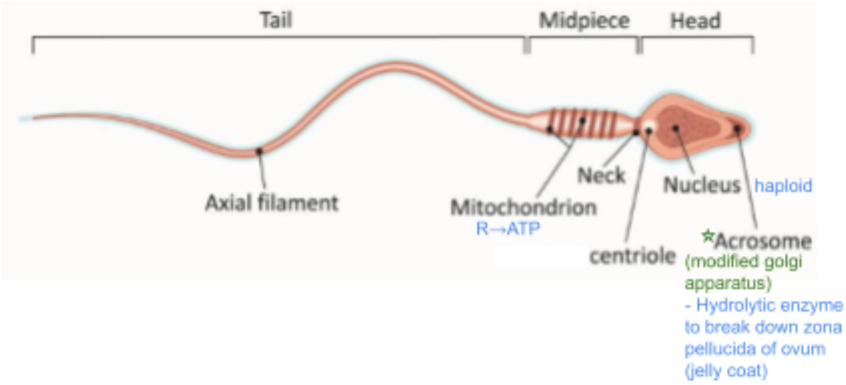


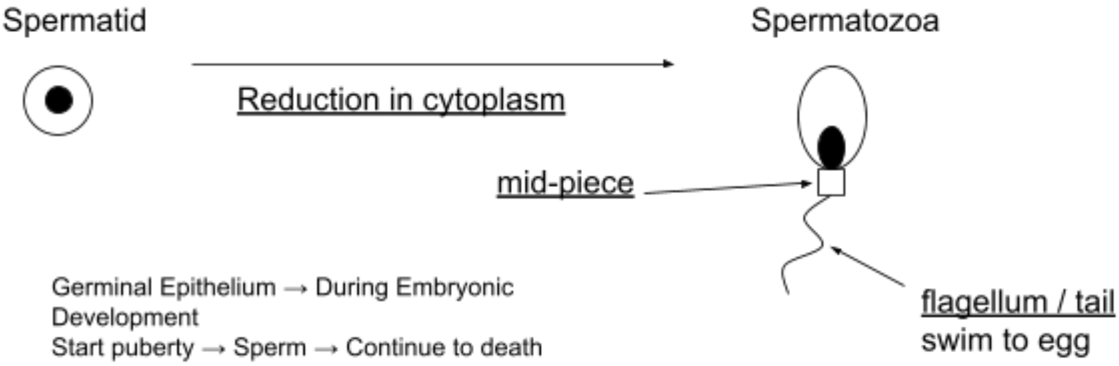
Female – Oogonia / oocyte develop inside follicles



Structures

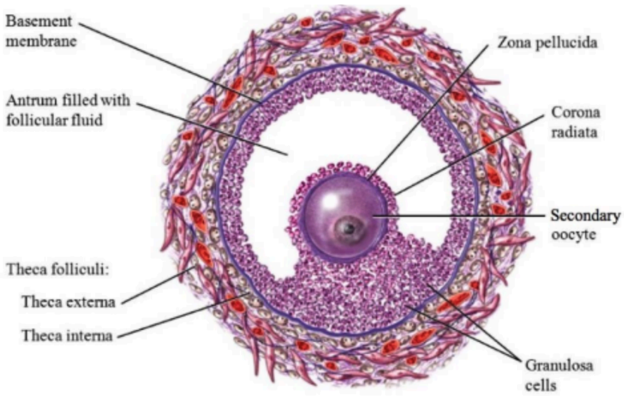
Spermatozoa



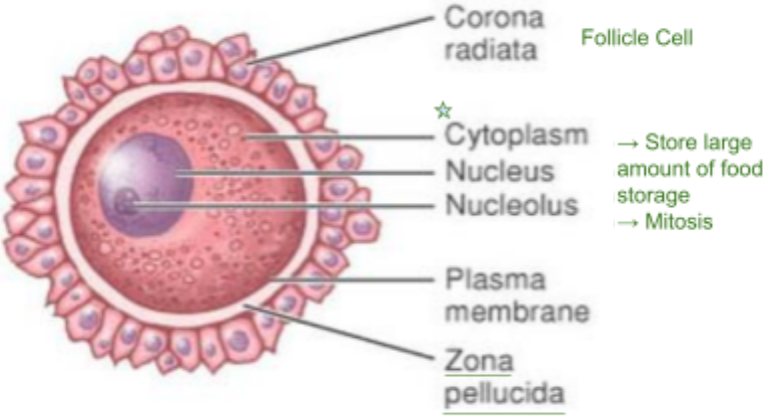


Germinal Epithelium → During Embryonic Development
Start puberty → Sperm → Continue to death

Graafian Follicle



Secondary Oocyte (After Ovulation)



Ovulation

- The release of the secondary oocyte from the Graafian follicle in an ovary.

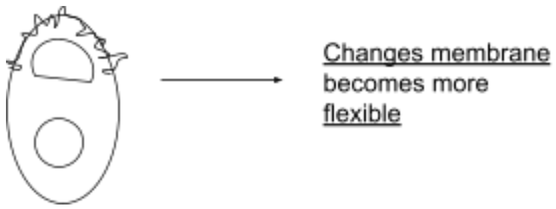
- Day 14 of menstrual cycle, secondary oocyte is released from ovary / Grafia Follicle
- Pressure in the antrum builds up and ruptures the Graafian follicle, forcing the secondary oocyte out.

After Ovulation

- Empty graafian follicle → corpus luteum
- Corpus luteum → release Oestrogen and Progesterone → Prevent further ovulation + Aid embryo development
- No fertilisation → slowly breakdown

Fertilisation

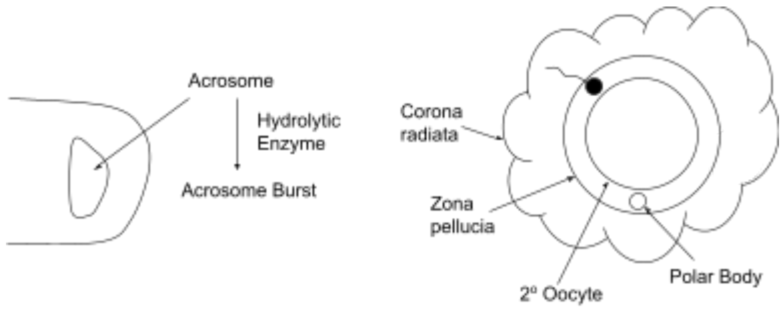
Capacitation



Reach the Secondary Oocyte

- Sperm respond to oocyte's chemo attractants, swim through cervix → uterus → oviduct
- Sperms remains viable for 2-5 days, Secondary oocyte remains viable for 24 hours

Acrosome Reaction

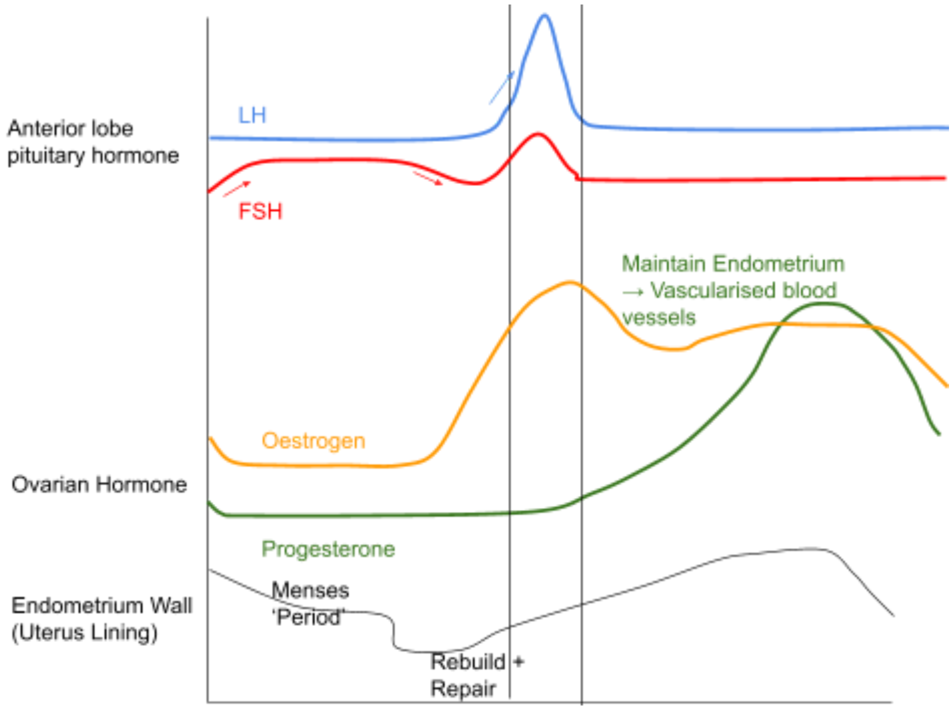


- Acrosome release enzymes to <u>digest cell of corona radiata</u>
- When sperm head contact with zona pellucida, <u>acrosome membrane ruptures</u> and release more enzymes including <u>protease</u> and acrosin which <u>hydrolyse zona pellucida</u>

Cortical Reaction

- Oocyte produce a fertilisation membrane <u>preventing polyspermy</u>
- When sperm attaches to secondary oocyte, oocyte smooth endoplasmic reticulum release calcium ions into cytoplasm
- Cortical granules fuse with membrane and release enzymes by exocytosis to destroy sperm binding site
- Zona pellucida modifies and expands and <u>harden</u> fertilisation membrane which does not allow other sperm to penetrate → ensure diploid condition can be restored
- Entry of sperm stimulates the oocyte to complete second meiotic cell division of ovum nucleus and expels second polar body
- Sperm chromosomes join the ovum chromosomes on cell equator → zygote

Menstrual Cycle



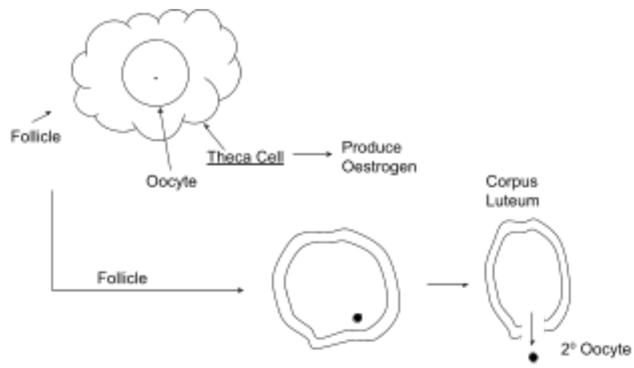
FSH (Anterior pituitary gland) → Stimulate follicle development in ovary

–ve effect on FSH ↑ ↓ +ve effect on oestrogen

Oestrogen (Ovary) → Cause endometrium to rebuild / repair

–ve effect on oestrogen ↑ ↓ +ve effect on LH

LH (Anterior Pituitary Gland) → Cause ovulation at Day 14



Progesterone (Ovary) → Maintains endometrium thicker, vascularised develop a good blood supply

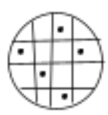
No Fertilisation

Corpus luteum degenerate

1. Progesterone production falls → Endometrium breaks down → Period

2. Oestrogen production falls → FSH production increases

Fertilisation



Blastocyst → Produce hCG → Stops corpus luteum degenerate → Carries on making Progesterone + Oestrogen

↙ Still a lining to embed into

↘ Inhibits FSH → No more ovulation

Hormone of Menstrual Cycle writeup

- FSH stimulate follicular development in the ovary
- Oestrogen is produced by follicular / thecal cell
- Oestrogen inhibits FSH production
- Oestrogen stimulates LH production
- *(LH initially stimulates oestrogen production)* LH then inhibits oestrogen production
- LH cause ovulation
- Oestrogen and progesterone are produced by the corpus luteum

Detailed Version of M.C. – WJEC

- FSH secreted by the anterior pituitary gland stimulates the maturation of a follicle and stimulates the production of oestrogen
- Following menstruation, the level of Oestrogen secreted by the developing follicle increases in the blood which triggers the repair of the endometrium; this inhibits FSH production and stimulates LH production;
- A high level of LH, secreted by the anterior pituitary, initiates ovulation causes Graafian follicle to develop into a corpus luteum
- Progesterone, secreted by the corpus luteum, causes further development of the endometrium prior to menstruation
- If implantation does not occur, falling FSH and LH levels cause the corpus luteum to degenerate, progesterone levels fall, the endometrium breaks down and is lost during menstruation; FSH secretion is no longer inhibited and another menstrual cycle is initiated.

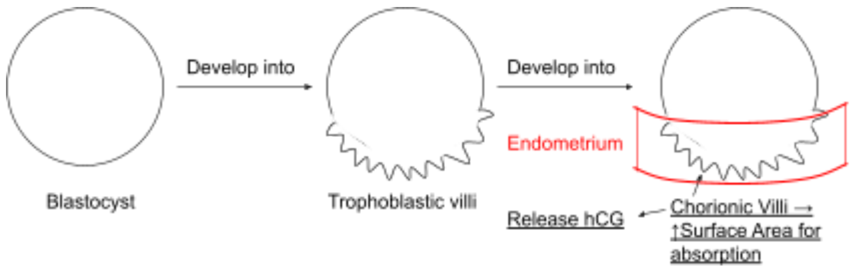
Note

- hCG maintains the corpus luteum and prevents it from degenerating
- Corpus luteum produce oestrogen + progesterone
- Progesterone: Maintains endometrium lining
- Oestrogen: inhibit FSH production + cause endometrium lining to repair
- *Oestrogen would have +ve effect on FSH when oestrogen levels are high (Legacy)* + –ve effect on oestrogen for most of the time
- Anterior pituitary gland produce FSH and LH
- FSH: Stimulate follicle development and cause an graffian follicle to mature inside ovary
- LH: Cause ovulation on day 14

IVF (Not in Spec)

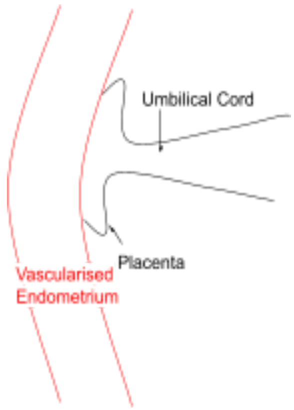
- Secondary oocytes are removed from a woman's ovaries and fertilised with sperm
- Implant Blastocyst (Day 18) → hCG
- hCG → Stop endometrium break down

Embryonic Membrane



Placenta

- Site of exchange
- Prevents foetal + maternal blood mixing



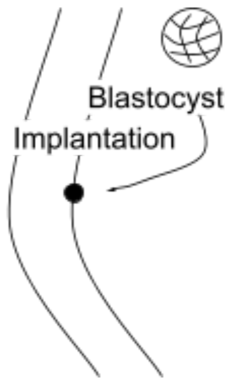
i) Prevents immune response by mother to foetal tissue → WBC would attack + kill foetus

ii) Maternal + foetal blood type are different → Possible death of one / both

iii) Barrier to pathogens (some can still cross HIV)

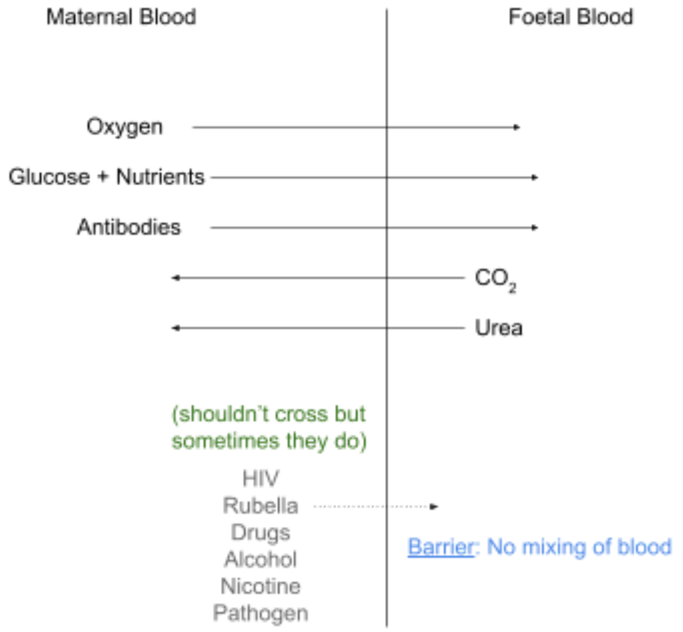
iv) Maternal high blood pressure would kill foetus

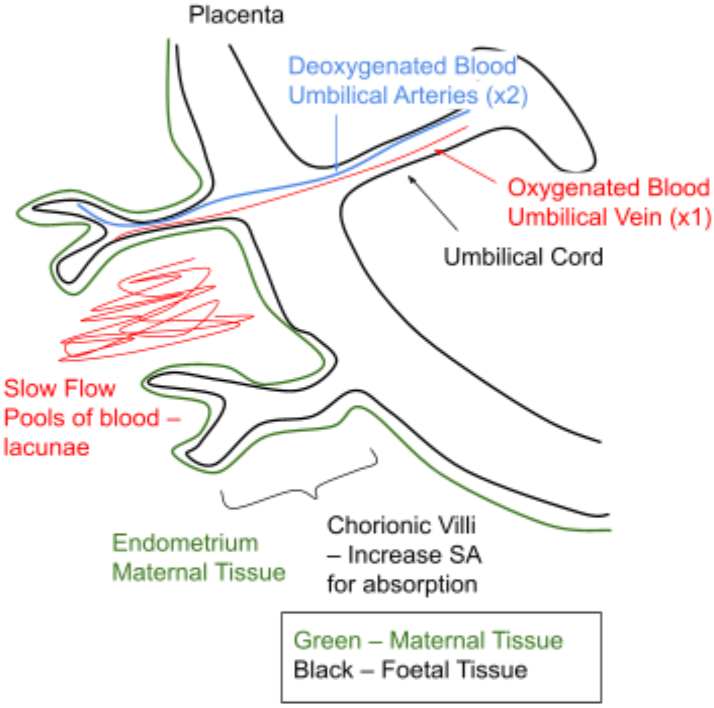
v) Barrier to toxins (not always) [eg: drugs]



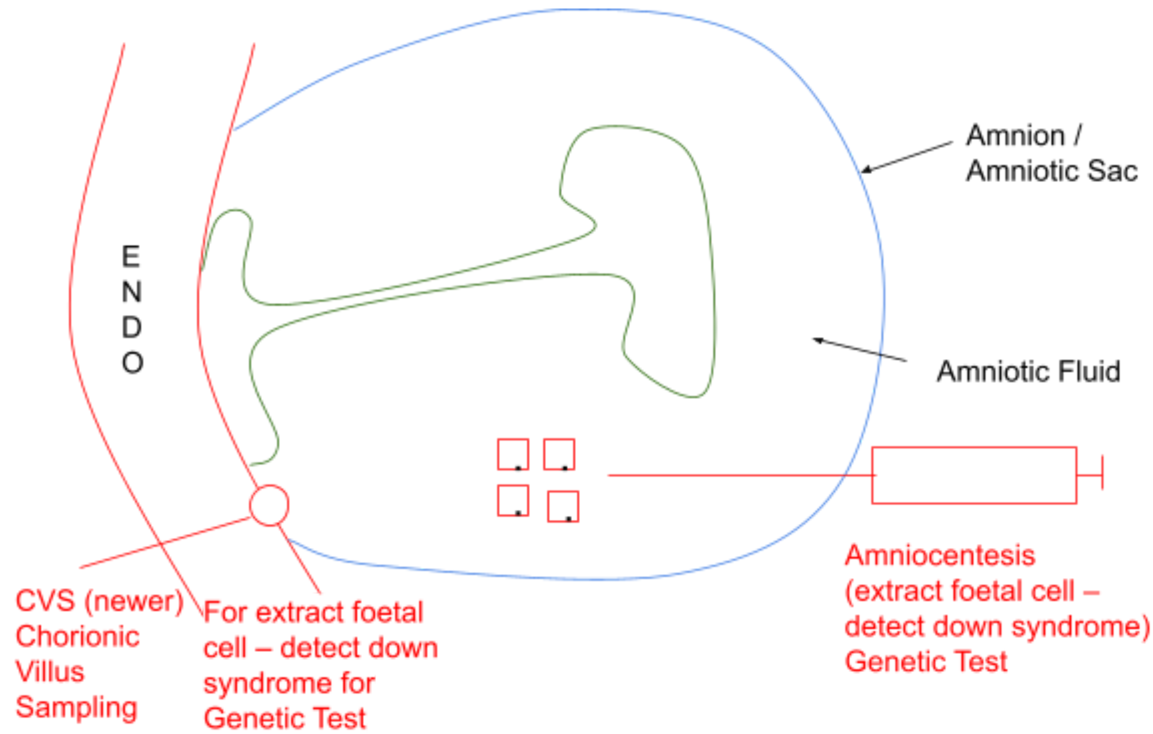
Vascularised Endometrium → Lots of blood vessels

Features of Placenta
- Short diffusion pathway
- Large surface area – chorionic villi + larger size – more than 30cm to be fully developed
- Excellent blood supply
    - Lacunae (pools of blood only in mother's blood pool) slow flow of blood → ↑ TIme for exchange
    - Counter current flow
- Short diffusion pathway *(2-3 cells + basement membrane)*





Amniotic Sac + Amniotic Fluid



Hormone in Giving Birth

TIMELINE

Menstrual Cycle | Fertilisation | Implantation

Corpus Luteum (ovary)

Maintains

Blastocyst → hCG

"pregnant" → cont.

Oestrogen from follicle before corpus luteum forms
→ (rebuilds endometrium)
→ Inhibits FSH

Production Continues

Oestrogen
→ Cause mammary gland growth (to produce after baby is born)
→ Cause uterine growth (room for developing foetus)

Inhibit FSH and LH (no more follicles / egg need)

Progesterone
→ maintains endometrium + cause vascularisation

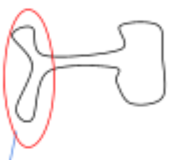
Production Continues

Progesterone
→ inhibits oxytocin production (prevents uterine contraction)

TIMELINE

cont. | 3½ → 9 months | After birth

Corpus luteum degenerates



Placenta

Oestrogen
→ Uterine growth
→ Mammary gland growth

Rise 1-2 weeks before birth

Inhibit FSH + LH
Inhibit prolactin (prevent milk production)

Progesterone
→ inhibit oxytocin

Fall 1-2 weeks before birth

+ve feedback

Prolactin Produced In anterior pituitary gland
→ Cause mammary glands to produce milk

Cause muscle contractions that deliver the milk

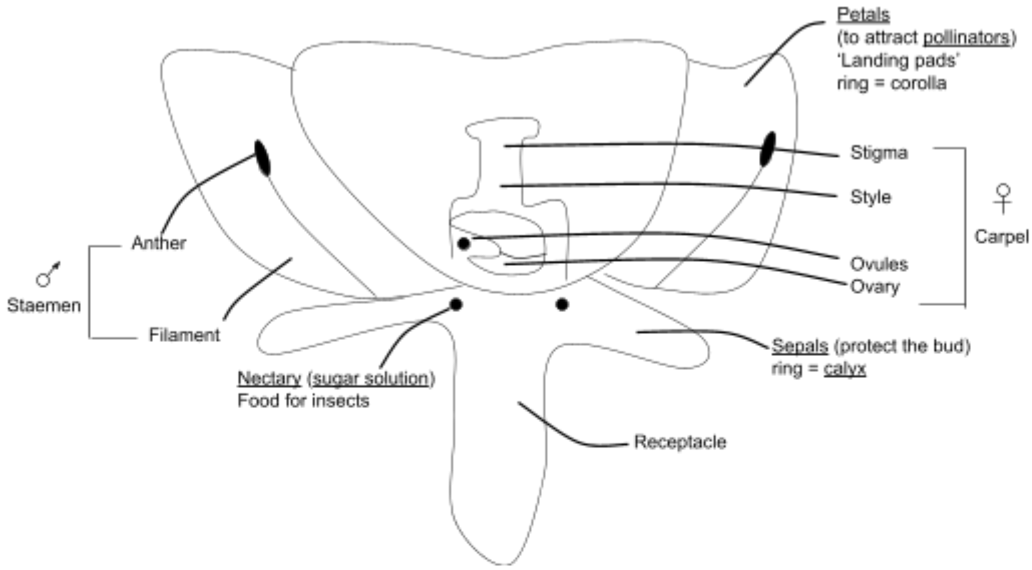
Oxytocin production increase in posterior pituitary gland
Contractions of myometrium / muscle in the uterus → Cause amniotic sac to burst → Birth

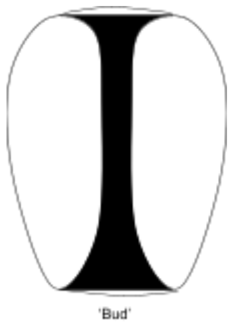
4.2 Sexual Reproduction in Plants

In WJEC we only talk about flowering plants (angiosperms) X Cornifers, X Ferns, X Mosses

Structure of a Insect Pollinated Flowers

Both sexes at the same time

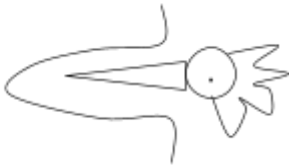




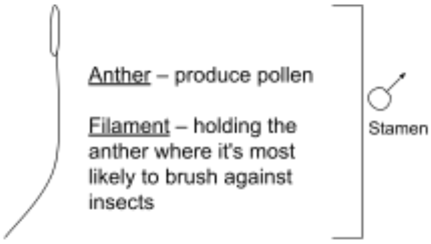
Structure	Function
Sepals	Protect flower when in bud (ring = calyx)
Receptacle	The thickened part of the stem, from which the flower grows
Corolla	A ring of colourful petals inside the sepal
Nectary	Found at the base of the flower Releases scented nectar to attract pollinators such as insects
Petals	In insect-pollinated plants they are colourful and have scents to attract insects.
Stamen	Male parts Made of an anther attached to a long filament
Anther	Pollen grains are produced inside 4 pollen sacs by meiosis
Filament	Contains vascular tissue, which transports mineral ions and water to the developing pollen grains
Carpel	Female parts Made of stigma, style and ovary
Ovules	Made inside the ovary Contain an egg cell formed my meiosis

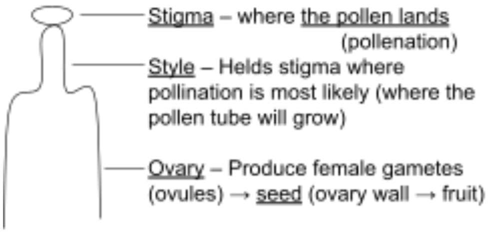
Flowers → Fruit + Seeds

Hummingbirds



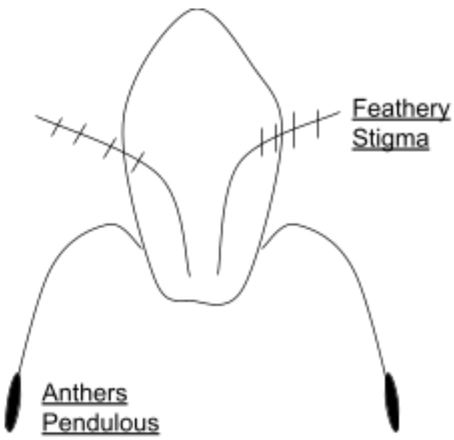
"Floral Formula" ⇒ relatives





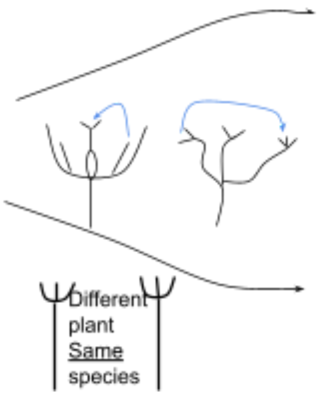
Part of an insect pollinated flowers		Part of a wind pollinated flower	
Petals	✗ →	NO!	
Stamen ♂ Anther / Filament	→	Stamen	Birch
Carpel ♀ Stigma / Style / Ovary	→	Carpel	Grass
Nectary	✗ →	NO!	Plantain
Sepal	→	Sepal	

Compositae



Pollination: When pollen lands on stigma

Pollination
When pollen lands on the stigma

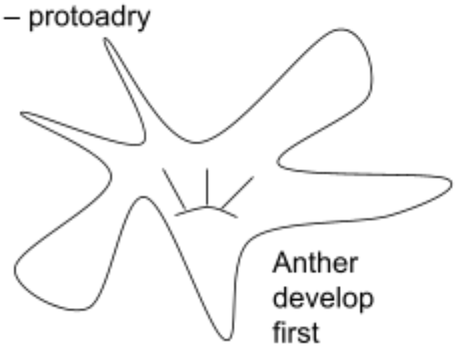


Self–pollination
Less genetic variation
When pollen from an anther lands on the stigma of same flower or different flower on the same plant.

Cross–pollination
Gives more genetic diversity
Variation

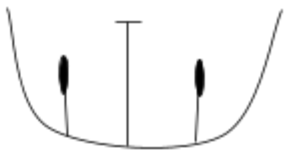
Ways of Preventing self–pollination
(allowing cross-pollination to take place)

Stamen and Stigma ripen at different times
Lily – protoadry



Anther
develop
first

Anther are below stigma so pollen cannot fall onto it



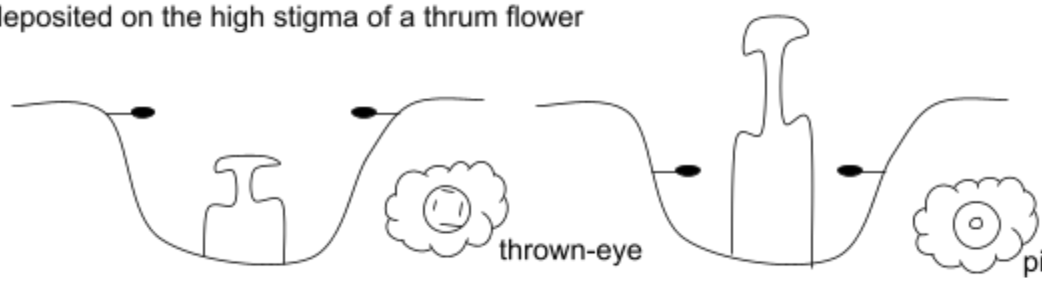
Separate male and female plant – holly

female

Self-incompatibility

Too genetically
similar

A bee landing on a pin flower picks up pollen at a low position, which is then deposited on the high stigma of a thrum flower



4.3 Inheritance

Definition

Gene	A sequence of DNA that codes for a polypeptide and which occupies a specific locus on a chromosome.
Allele	A variant of a gene; it has a different base sequence and will result in the production of a different polypeptide.
Dominant	An allele that will be expressed in the phenotype if only one copy is present.
Recessive	An allele that will only be expressed in the phenotype if two identical alleles are present.
Phenotype	The physical characteristics of an organism; a result of the expression of its genes.
Genotype	The genetic make-up of an organism; the specific allele combination for a particular gene.
Locus	The location of a gene on a chromosome.
Mutation	A change in the amount or structure of DNA within a cell.
Homozygous	An individual with two identical alleles for a gene.
Heterozygous	An individual with two different alleles for a gene.

Test Cross / Back Cross – GCSE Example

A guinea pig with black fur (homozygous) is crossed with a white-haired (homozygous → 2 copies of the same alle in a genotype [genetic combination]) guinea pig. All of the F_1 has black hair. Using suitable symbols show this cross and predict the appearance of the F_2 if the F_1 are selfed.

- Allele – Variant of a gene
- Recessive – Will only be express if phototype if two copies are present
- Dominant (black) – Will be expressed in the phenotype [physical appearance] if one copy is present

Parental Phenotype black white

Parental Genotype BB bb

Gametes B B b b
B = black
B = white

	b	b
B	Bb	Bb
B	Bb	Bb

One copy of each allele (2 different alleles)

F_1 → all black, heterozygous

	B	b
B	BB	Bb
b	Bb	bb

3 black
1 white

Homozygous Dominant (black) – BB
Heterozygous (black) – Bb
Heterozygous Recessive (white) – bb

Reason on why in experiment it might not be exactly 3:1 ratio (eg 3.009:1)

- Random fertilisation
- Sample size too small

Mendel's First Law

- Two alleles of a gene separate from its partner and moves into different gametes
- Each gamete can only receive on allele from each pair

Codominance

2 alleles both dominance
WJEC Examples: Cattle / Snapdragon Anthurium

Cattle

Red x White
Roan (RW)

Red RR   White WW

F1: RW   RW

F2:

	R	W
R	RR	RW
W	RW	WW

1:2:1
red:roan:white

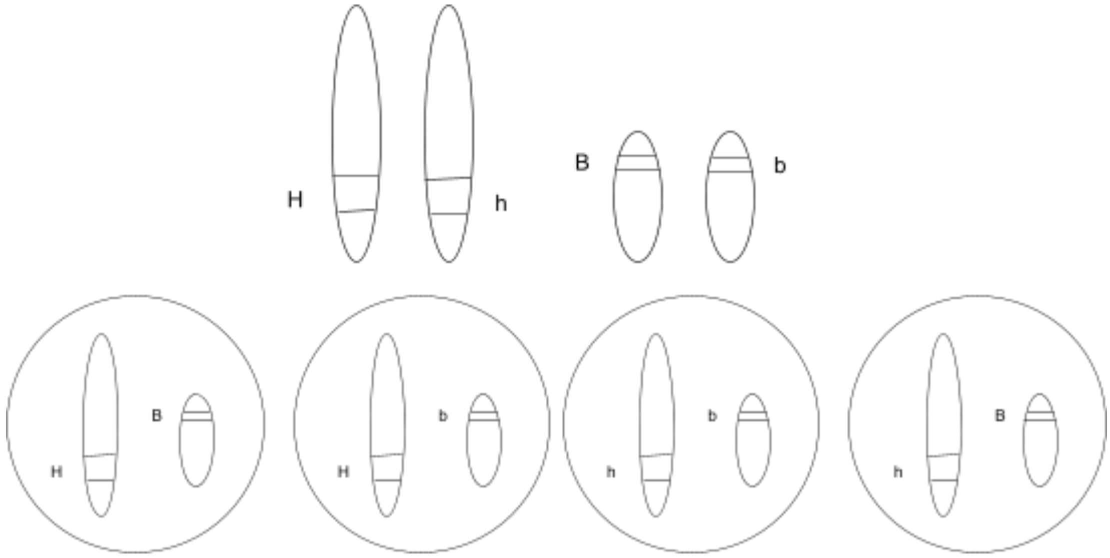
Blood

Blood Type	Genotypes
A	I^AI^B / I^Ai
B	I^BI^B / I^Bi
AB	I^AI^B
O	ii

Chi^2

- Null Hypothesis: There is no statistically significant difference between the means of [height / mass / description] of the two samples and that any difference is due to chance.
- Degree of Freedom: Number of Sample 1 + Number of Sample 2 – Number of Samples
- Probability: 5% (or 0.05)
- Find critical value
- If t value > critical value → reject null hypothesis → not due to chance → due to other factors: (for example lack of Mg2+)
- If t value < critical value → accept null hypothesis → any difference is due to chance

Dihybrid Inheritance



Assumption: Two traits of genes are located on different homologous chromosomes

Example with Peas

Characteristic 1: Color of peas (Yellow **Y** / Green **y**)
Characteristic 2: Texture of peas (Round **R** / Wrinkled **r**)

Pure Bred Parents

Yellow, Round (YY RR) + Green, Wrinkled (yy rr)
→ F_1 all heterozygous Yy Rr which is yellow round

F_2 Generation

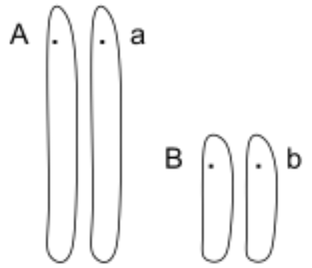
	YR	Yr	yR	yr
YR	YY RR	YY Rr	Yy RR	Yy Rr
Yr	YY Rr	YY rr	Yy Rr	Yy rr
yR	Yy RR	Yy Rr	yy RR	yy Rr
yr	Yy Rr	Yy rr	yy Rr	yy rr

Yellow and Round (2D) – 9
Green and Round (1R1D) – 3
Yellow and Wrinkle (1D1R) – 3
Green and Wrinkle (2R) – 1
Two dihybrid heterozygous breed will form 9:3:3:1 pattern

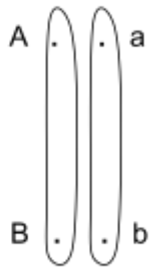
Linkage

23 Pair Chromosomes

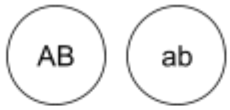
Dihybrid



Linkage



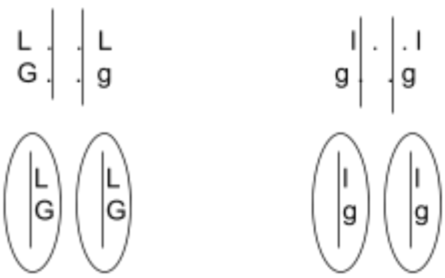
2 genes are on same chromosome → go into a gamete + be inherited together



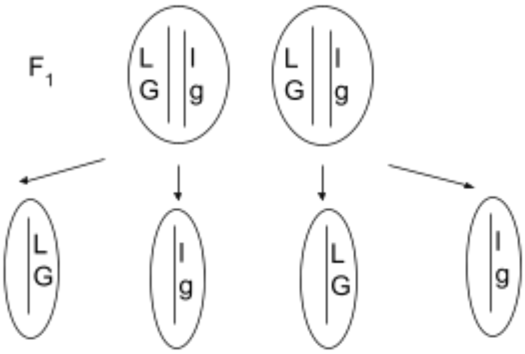
No Ab or aB

L = long
l = short
G = grey
g = white

Long Grey x Long Grey
LLGG llgg



F1



	LG	lg
LG	LLGG	LlGg
lg	LlGg	llgg

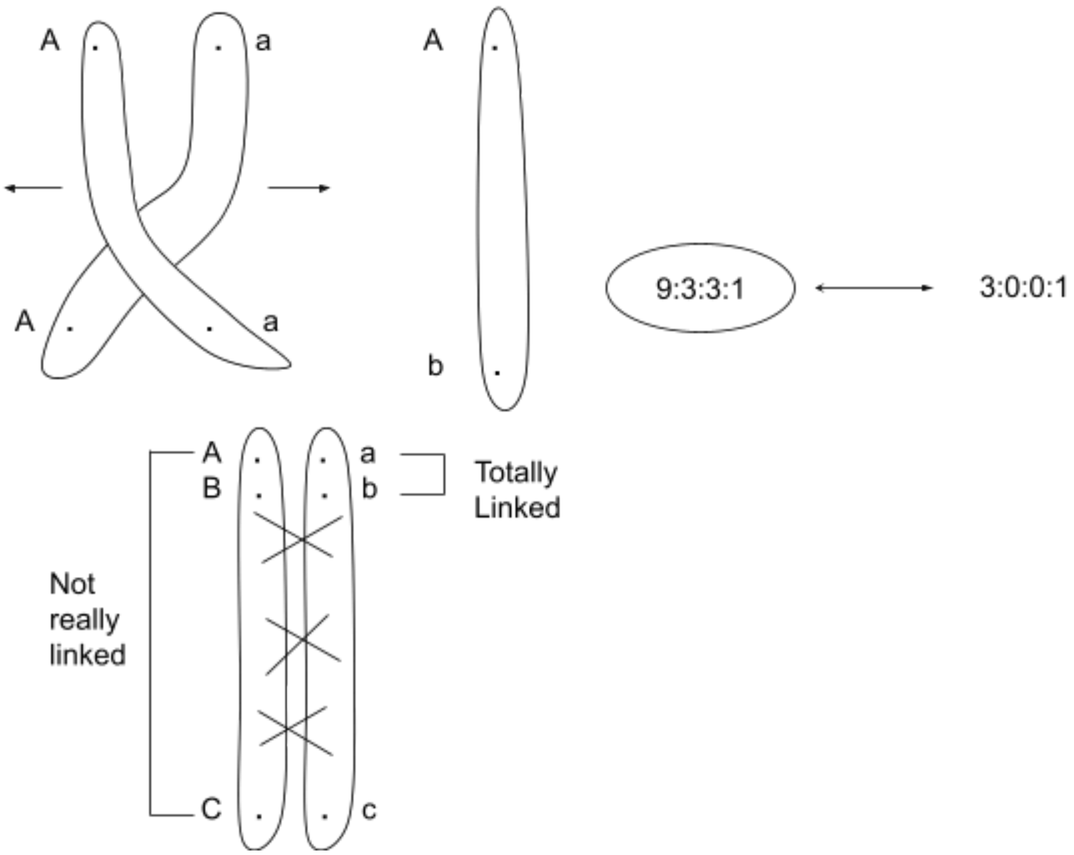
Long Grey : Short White
3 : 1

No Recombinants
Missing Long White , Short Grey

Ratio: 3:1 ← Aa x Aa (hetero)
Ratio: 1:2:1 ← RW x RW (codominance)
Achondroplasia → 'Dwarfism'

Aa x Aa
AA Aa Aa aa
lethal

Prophase I



If dihybrid is not 9:3:3:1

- Recombinds are less frequent → genes are <u>linked</u> → Genes are on the same chromosome + Therefore will be inherited together

There are still some recombinants present

- Crossing over during prophase I of meiosis

Chi Squared Significance → Due to linkage

Sex Determination

Female
XX

Male
XY

	X	Y
X	XX	XY
X	XX	XY

44 Chromosomes
2 Sex Chromosomes
Not a sex chromosome = <u>autosome</u>

Sex Linkage

More common in 1 sex (WJEC: X–linked) which is male
As only 1 chromosome (X) is required to show the genotype

- Haemophilia
- Colour Blind
- DMD*

As most Sex Linked are <u>recessive</u>
Example:

This example:
H = normal
h = haemophilia

or
B = normal
b = colour blind

or
D = normal
d = DND

Normal
Male
X^HY

X^H Y

Carrier
Female
X^HX^h

X^H X^h

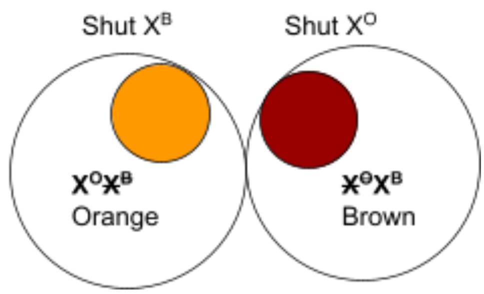
	X^H	Y
X^H	X^HX^H	X^HY
X^h	X^HX^h	X^hY

2 normal female (1 carrier)
1 normal male
1 haemophilia male

Lyonization (Extension)

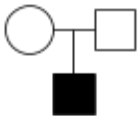
- Occurs in female
- As Female have 2 active X chromosome
- Shut 1 down / inactivate it at random → ‘barr body’
- Example: Tortoishell Cat

Tortoiseshell Cat → Female
X^oX^B → Codominance

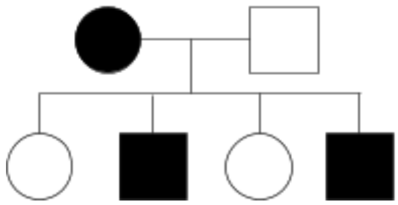


Pedigree Diagram

Key:
If both parent dont have trait but offspring have → Recessive (parents are carrier)

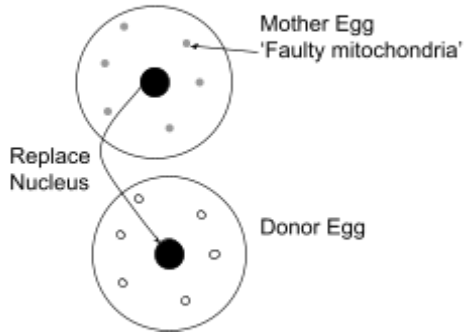


X Linked Recessive
(look at mother suffer + son suffer)



3 Person Baby

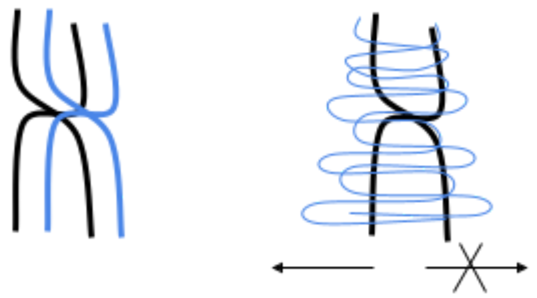
- *Mitochondria come from egg in a zygote
- If mother have faulty mitochondria → Replace the haploid nucleus from the defective mother's egg to the donor's egg
- Sperm fertilise with the donor's egg (which has healthy mitochondria) → genetically still 99% similar (only mitochondria is from donor)



Mutation

- Change in DNA
    - Change in base sequence → Different amino acid → No-funcitonal / different protein → During DNA Replication
    - Change in the amount of DNA → During Cell Division → (eg Down Syndrome)

Change in amount of DNA
Occurs in Prophase I or Anaphase I/II



'non-disjunction'

Change in base sequence → Gene mutation (point mutation)

Base Addition or Deletion
→ 'Frame Shift' → every codon / triplet is wrong

Coding
Strand DNA

Base Substitution
→ Wrong amino acid → 1° structure wrong

May affect 2° / 3°

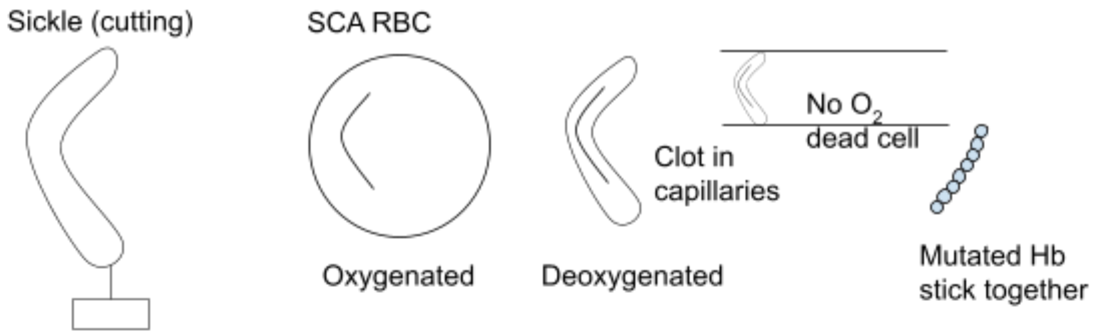
If 2 amino acid similar
→ very little effect

→ Degenerate code → Same amino acid
'Hidden mutation'

→ STOP! code beyond this won't be read

Sickle Cell Anaemia / Thalassaemia (Base Sequence)

- Base substitution (Point Gene Mutation)
- B-chain of Hb molecule



Hb^A = normal
Hb^S = sickle cell

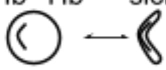
co-dominance

$Hb^A Hb^A$ = normal → malaria

$Hb^A Hb^S$ = sickle cell trait → do not get malaria

Don't get SCD

$Hb^S Hb^S$ = sickle cell disease → do not get malaria



Common sub saharan Africa

	Hb^A	Hb^S
Hb^A	malaria	😃
Hb^S	😃	SCD

Chromosome Mutation

- Prophase I
- Anaphase I / II – Non disjunction
- Example Down Syndrome / 'cri-du-chat'

Down Syndrome (Triosomy 21)

- 3 copies of chromosome 21 – non-disjunction
- (also part of chromosome 21 extra – attracted to another chromosome)
- Wrong number of chromosome → FATAL (almost always)
- –1 / +1 Chromosome 'aneuplody

Polyploidy

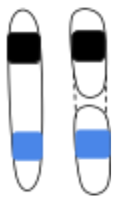
14 → Wheat 15 ← 16

Endomitosis x2

Fertile — 30 — Multiple sets of chromosomes

Apple = 6n / 8n
Kiwi = 8n
Strawberry = 6n - 10n

Human = 46
Chimpanzee = 48



Cancer

Mutation

Germ Line Mutation — Germinal Epithelium in Testes / Ovaries → Pass mutation on

Body Cell Somatic Cell

Mutation leads to cancer

before
Protooncogene

Could lead to uncontrolled cell division

Tumour Suppressor Gene

Stop cell division

Don't hit the break

<u>Oncogene</u>
Cell division in an uncontrollable way

Accelerator
Cancer

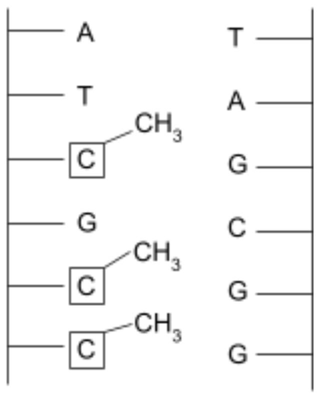
<u>Metastasis</u>

Epigenetics

- Differences / changes in gene expression <u>WITHOUT</u> changing the structure of DNA
- Gene On / Off

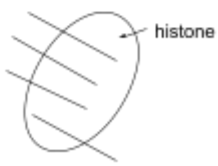
1. Cytosine in the gene — Methylate Cytosine Boxes → 'off'

<u>Turn Off</u>
Lower the expression of the gene
Reduce its use in protein synthesis



2. – acetyl group of histone → DNA is held more <u>loosely</u> → easier to transcribe 'read' → 'ON'

   – remove acetyl group from histone → DNA is bound <u>more tightly</u> → not transcribe → 'OFF'



4.4 Variation and Evolution

Variation

The differences between organisms of the same species

Variation could be due to
- Genetic differences – different alles
- Epigenetic modification – control gene expression
- Environmental differences

Heritable Variation

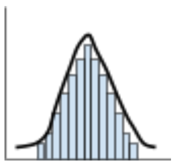
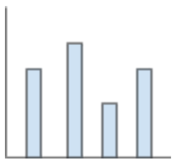
Asexual Reproduction
- Random mutation

Sexual Reproduction
- Crossing over during prophase I of meiosis
- Independent assortment of chromosomes during metaphase I
- Independent assortment of chromatids during metaphase II
- Random fertilisation

Non heritable Variation

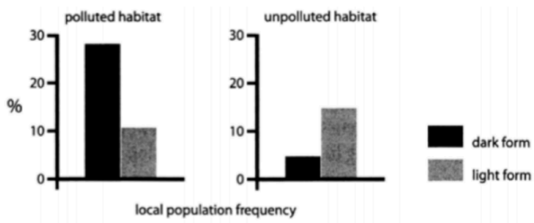
- Due to environmental changes
- Cannot be inherited by the offspring

Categories of Variation

	Continuous Variation	**Discontinuous Variation**
Properties	- No distinct categories - No limit on value - Quantitative	- Discrete categories - No in between categories - Qualitative
Examples	Sea snail shell size	Blood Type
Graph		
Controlled by	Polygenic (controlled by multiple genes for phenotype) + Environmental	One gene

Peppered Moth – Discontinuous Variation

- Peppered moth can only be light or dark
- At polluted habitat, dark coloured are harder to spot → more common



Competition

Same	Difference
Intarspecific	Interspecific

- Limited resources availability (eg: food) → limit population growth around carrying capacity

Selection Pressure

An environmental factor that can alter the frequency of alleles in a population, when it is limiting.

Alleles and the Environment

- If phenotype is advantages, it is more likely to be selected for
- Allele frequency increase over time

Gene Pool

- All of the alles present in a population at a given time

Hardy–Weinberg Principle

In a population where there are 2 possible alles, the frequency of dominant and recessive alles will be 100% (All are either BB, Bb, bb)

$$\mathbf{p + q = 1}$$

$$\mathbf{p^2 + 2pq + q^2 = 1}$$

p = frequency of dominant allele (A)
q = frequency of recessive allele (a)

p^2 = frequency of homozygous dominant (AA)
2pq = frequency of heterozygous (Aa)
q^2 = frequency of homozygous recessive (aa)

Conditions

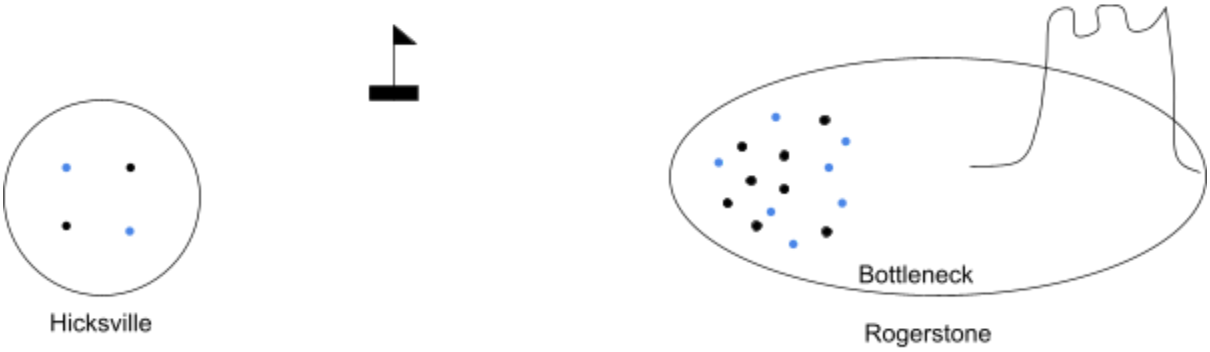
- A large population
- No selection for or against phenotype
- Random mating throughout the population
- No mutations
- The population is isolated ie: No immigration or emigration

Evolution

Change in frequency of allele over time // Change on the average phenotype of a population

Genetic Drift

- RANDOM
- Chance variation in allele frequency in a population



Founder Effect

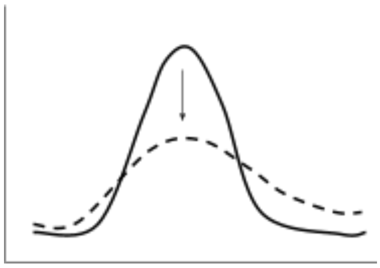
- A few individuals from a population start a new population with a different allele frequency than the original population
- Small number of individual become isolated and start a new population
- Founder members are a small sample of the original population → different allele frequency from original population
- Founder population has a small gene pool and lower diversity

Bottleneck Effect

- Sharp reduction in a population and a sharp reduction in gene pool due to earthquake / fire
- An outside forces destroy most of a population
- Increase inbreeding due to reduced gene pool → Little genetic diversity left in gene pool
- Increase expression of recessive trait
- Eventually there will be a large genetically similar population of organisms

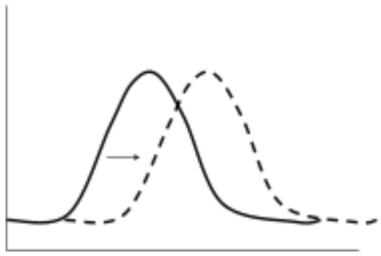
Natural Selection

Stabilising Selection



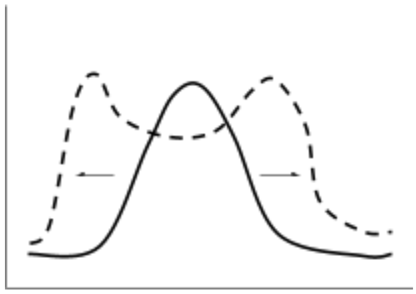
- Natural selection that favours the proportion of the average phenotype in a population
- It selected against the extreme population → reduce variability

Directional



- Selection of one extreme of a phenotype
- Example: Peppered moth change colour from light to dark colour

Dispersive



- Average phenotype is selected against
- Bimodal peak

T Test

* Significance * Mean
H_0: There is no statistically significance between the mean values of the shell size of population a and population b.

Example: Sea Snail Shell Size
We collected 15 samples from habitat a and 15 from b
Our calculated t value was 2.55
DoF = 15 + 15 – 2 = 28
Confidence Limit = 0.05
Critical Value = 1.701

Reject H_0 ∵ As out calculated value of 2.55 is above the critical value of 1.701 (with 28 dof and a confidence level of p=0.05) we must reject null hypothesis, and accept that there is a significant difference between the mean shell size in the exposed and sheltered shore line.

Isolation and Speciation

Reasons to tell whether organism can interbreed and produce fertile offspring:
- Organisms have not reached sexual maturity
- Organism will not mate
- Very long reproductive cycles
- Organisms reproduce a sexually

Speciation

- Pre-zygotic: Gametes are prevented from fertilising each other, so <u>a zygote does not form</u>
- Post-zygotic: Gametes can fertilise each other <u>to form a **zygote**</u>. However the organism formed is a hybrid and is sterile **(chromosomes cannot pair up to in meiosis to form games)**
- **<u>Deme</u>**: A subgroup within a population that may breed more often with each other than with the rest of the population.
- Reproductive Isolation: The prevention of reproduction and gene flow between breeding groups within a species.

Pre-zygotic

Geographical Isolation – Only Allopathic Speciation (Different Geological Location)

- <u>Physical barrier</u> → Prevent <u>flow of allelesPollen grain does not germinate on the stigma</u>
- Different selection pressure → Different phenotype selected for → Allele frequency changes

Behavioral Isolation – Sympatric Speciation (Same Geological Location)

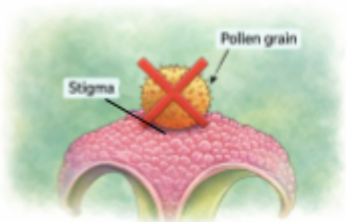
- <u>Different courtship display or mating call</u>
- Not recognize each other as same species → Prevent from breeding each other
- Prevent them breeding with each other → Mutations → Formation of different species

Morphological Isolation – Sympatric Speciation

- <u>Different body form</u>
- 2 population may be compactable
- Prevent alleles from 2 group mixing

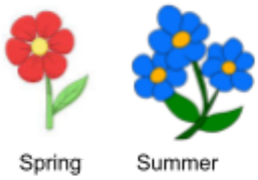
Genetic Isolation – Sympatric Speciation

- <u>Gamete</u> from different population are <u>prevented from coming into contact with each other</u>



Seasonal / Temporal Isolation (Plant)

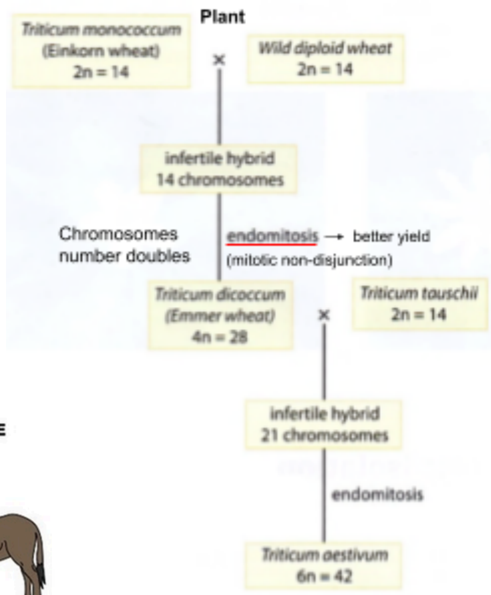
- Reproduce organs mature in different time of year → Prevent breeding



Post Zygotic

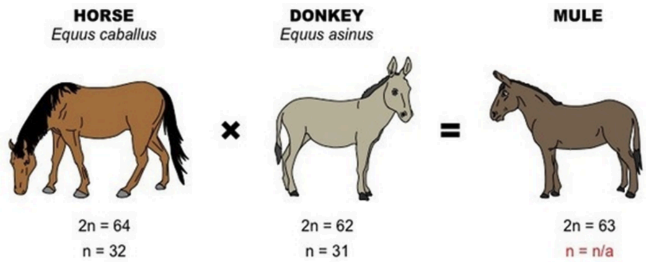
Hybrid Inviability (Embryo development failed to occur)

- Inviability: **Cannot survive**
- Gametes can fertilise other to form zygote
- <u>Embryo failed to form</u>



Hybrid Sterility (Polyploidy)

- Gametes can fertilise together to form offspring
- Fertile offspring is sterile
- Chromosomes from different species are not homologous
- <u>Chromosomes cannot pair up in prophase I of meiosis</u>
- Meiosis cannot continue and gametes cannot form



Hybrid Breakdown (Plants [less common])

- F1 is fertile but F2 is sterile
- Example: Nuclear genes vs Mitochondria / Chloroplast genes

Summary Table (Learn This)

	Type of isolation	Sympatric or allopatric	Example
Pre-zygotic	Geographical	Allopatric	Sticky cinquefoil
	Behavioural	Sympatric	Grasshopper
	Morphological	Sympatric	Insects
	Gametic	Sympatric	Sea urchin
	Seasonal	Sympatric	Buttercup and lesser celandine
Post-zygotic	Hybrid inviability	Sympatric	Northern leopard frog and wood frog
	Hybrid sterility	Sympatric	Mule
	Hybrid breakdown	Sympatric	Cotton

Darwin Evolution

☆ – Gene Flow
☆ – Alleles passing over

Darwin Deduced

- Survival of fittest
- Survive and reproduce to pass on allele
- New species forms when many changes
- Environmental changes, natural selection is a continuous process

Natural Selection (A Level Version)

- <u>Variation</u>
- <u>Competition</u> of resource
- Advantageous alleles are selected for
- <u>Survive to adulthood</u> and reproduce
- Alleles passed onto offspring for many generations
- <u>Alleles increase in frequency</u> in a gene pool
- Organism with alleles which have lower tolerance will die and cannot pass on gene
- New species formed

Tips:

In A Level, use the word <u>Gene Flow</u> and <u>Allele Frequency</u> in terms of natural selection to gain more mark

4.5 Application of Reproduction & Genetics

Genomics

Study of the structure, function, evolution and mapping of genomes

Human Genome Project
- Sequence all genes in the human genome and identify which chromosome each is on
- Sequencing Method: Sanger Sequencing

The 1000K Project
- Sequence 100,000 different human genomes
- Sequencing Method: Next Generation Sequencing (NGS)
- Advantage: Faster and Cheaper, Medical Research for Healthcare

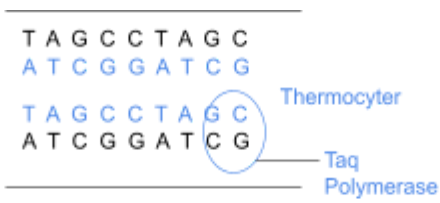
Genetic Sequencing – Determines the exact base sequence of an entire genome,

Genetic Fingerprinting

- Short Tandem Repeats (STR) produce variation on individuals

Why? To provide lasting protection
Who? Everyone potential at risk
Cancer? Cost / ownership of data

PCR (Polymerase Chain Reaction)



- Taq (DNA) Polymerase
- ATCG
- Primer (x2)

1. Denature DNA by rising temperature to 95° to break H bonds
2. Lower the temperature to 55°C so the primers can <u>anneal</u>
3. Raise the temperature to 70°C for Taq polymerase to <u>replicate</u>

Taq polymerase use instead of DNA polymerase → prevent denature of active site

Limitation:
- <u>Contamination of DNA</u> in machine → as DNA fragment is small
- High error rate
- Limited DNA fragment site
- Sensitivity to inhibitors
- Amplification limits

4.A Immunology and Disease

Definition

Pathogen	an organism that causes damage to its host.
Infections	a disease that may be passed or transmitted from one individual to another.
Carrier	a person who shows no symptoms when infected by a disease organism but can pass the disease on to another individual.
Disease reservoir	where a pathogen is normally found; this may be in humans or another animal and may be a source of infection.
Endemic	a disease, which is always present at low levels in an area.
Epidemic	where there is a significant increase in the usual number of cases of a disease often associated with a rapid spread.
Pandemic	an epidemic occurring worldwide, or over a very wide area, crossing international boundaries and usually affecting a large number of people.
Vaccine	uses non-pathogenic forms, products or antigens of micro-organisms to stimulate an immune response which confers protection against subsequent infection.
Antibiotic	substances produced by microorganisms which affect the growth of other microorganisms.
Antibiotic Resistance	where a microorganism, which should be affected by an antibiotic, is no longer susceptible to it.
Vector	a living organism which transfers a disease from one individual to another.
Toxin	is a chemical produced by a microorganism which causes damage to its host.
Antigenic Types	organisms with the same or very similar antigens on the surface. Such types are sub groups or strains of a microbial species which may be used to trace infections. They are usually identified by using antibodies from serum.
Antigen	A protein / marker molecule that initiates / triggers an immune response.

Infectious Disease

Cholera	Type of organism	Gram Negative Comma Shape Bacterium: Vibrio cholerae
	Mode of transmission	Contaminated water
	Tissues affected	Gut lining → watery diarrhoea → severe dehydration
	Treatment	Rehydration (antibiotic treatment is possible)
	Prevention / Control methods	Treatment of water, good hygiene and provision of clean drinking water Vaccine may provide temporary protection
Tuberculosis	Type of organism	Bacillus shaped bacterium: Mycobacterium tuberculosis
	Source of infection / Mode of transmission	Aerosol transmission through droplets.
	Tissues affected	Lungs → tubercles → Chest pain and Phlegm and blood with cough. Lymph nodes.
	Treatment	Antibiotics over long durations
	Control methods	Vaccines give around 75% protection for 15 years.
Smallpox	Type of organism	Virus: Variola major
	Source of infection / Mode of transmission	Inhaled or transmitted in saliva or close contact with infected person
	Tissues affected	Enters blood vessels in skin, mouth and throat High fever + Fatigue Case rash and blisters → might suffer blindness and limb deformities
	Treatment	Given fluid, drug to control fever and pain Antibiotics to control bacterial infection
	Prevention / Control methods	Given fluids, drugs to control fever and pain. Vaccine (Cowpox virus) Antibiotics are ineffective against virus (so cannot control flu) [Antibiotics can act of cell wall of bacteria and affect protein synthesis by bacteria cell Virus does not have cell wall]

Influenza	Type of organism	Subgroup of virus: Influenza A, B and C
	Mode of transmission	Aerosol transmission.
	Tissues affected	Infects cells of the lining of the upper respiratory track Neuroaminidase on the virus' surface breaks down protective layers Secondary Infection: Pneumonia ← Antibiotic Treatment
	Treatment	Aspirin relieves symptoms. <u>Antibiotics are ineffective</u>
	Prevention / Control methods	Wash hands, quarantine Vulnerable people can receive immunisation Virus works but as the virus can mutate quickly, vaccines quickly become ineffective
Malaria	Type of organism	Protoctistan: Plasmodium Two deadliest species: Plasmodium falciparum and Plasmodium vivax
	Source of infection / Mode of transmission	Parasite multiplies in the liver before emerging to invade red blood cells. They emerge from the cells and release toxins in the blood.
	Tissues affected	Head aches, fevers Blood vessels and kidneys can be damaged.
	Treatment	Drugs: Quinine, Atovaquone (multidrug approach) **Quinine**: Disrupt digestion of haemoglobin + Produce a toxic derivative which kills Plasmodium **Atovaquone**: Affect ETC in Plasmodium mitochondria + unable to respire for plasmodium
	Prevention / Control methods	**Responding to mosquito behavior:** Sleep under nets Drain or cover Stagent water **Biological control:** Male mosquito sterilise with x-ray Fish introduce into water

Influenza

Surface antigens – Envelope surface contains two types of antigen proteins. H and N
Haemagglutinin (H) – Role for virus entering the host cell.
Neuraminidase (N) – Role for virus leaving the host cell.

Antigenic drift vs Antigenic shift

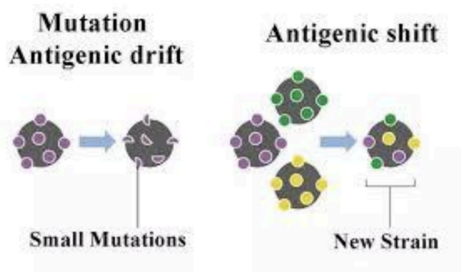
- As there are no RNA proofreading enzyme, each round of replication will have mutation
- When there is an antigenic variation, eg: Mutation
    - Antibodies have to produced again, as memory cells cannot recognise mutated antigen
    - Primary response each time

Antigenic Drift

- Accumulation of mutation within gene that code for antibody–binding site (antigens)

Antigenic Shift

- More major change in influenza virus
- Occurs when human flu virus crosses with a flu which affects animals
- During mutation, they shift to create a new subtype which is different from any seen in human before
- Likely to cause new epidemic

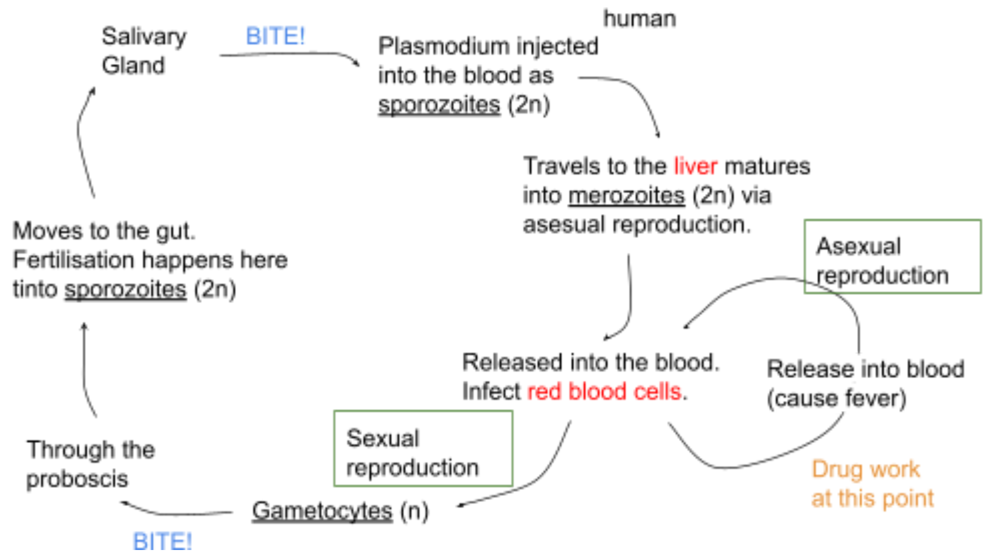


Mutation of Virus

- High mutation rate of virus → antigen changes
- Many strains / serotypes (types) of rhinovirus
- Vaccines need to contain antigen from all strains to provide immunity
- Each antigen must be immunogenic

Malaria

Plasmodium Life Cycle



Plasmodium life cycle in Human

1. Infected mosquito takes blood from another human by injecting saliva – anticoagulant
2. Plasmodium parasite also inject into the blood in the form: Sporozoites
3. Sporozoites travel to liver and reproduce asexually to form: merozoites
4. Merozoites burst out of liver cell into bloodstream
5. Merozoites infect red blood cell, merozoites are produced asexually and more are released from RBC to blood stream → fever
6. Merozoites also can become gametocytes

Plasmodium life cycle in Mosquito

1. Mosquito take blood from infected human in the form of Plasmodium gametocytes
2. Gametocyte travel to gut where fertilisation forms sporozoites
3. Sporozoites migrate from mosquito to salivary glands

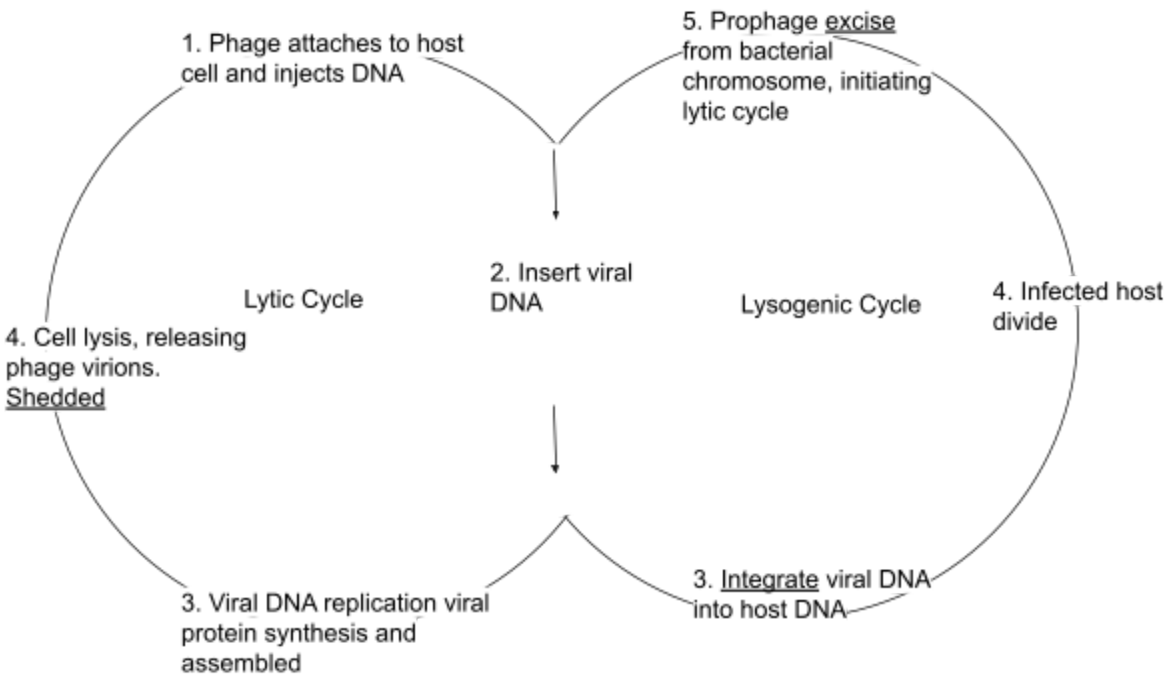
Vaccine

- Live but weakening organism which has been modified
- BCG vaccine

Virus pathogenicity & reproduction

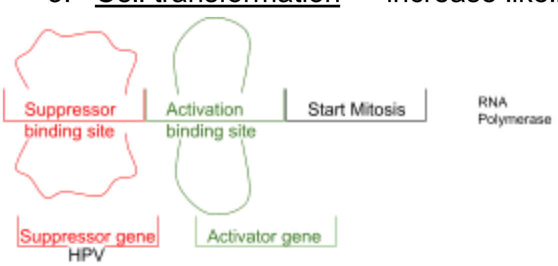
Mutualistic bacteria / fungi – Symbiosis between two organisms where both benefits.
Virion – The complete, infective form of a virus outside a host cell, with a core of RNA and a capsidum.
Virus – Reproduce inside host cell → host cell lysis to release many new virus
Virus can infect animal, plant and bacteria

Lytic Cycle + Lysogenic Cycle



Viral Infectious

1. Lysis of cell: Bacteriophage infect → Inflammation → Rinovius → Cause cell lysis
2. Produce toxin: Viral component produce toxin effect → chromosome fusion → cell fusion → inhibit DNA / RNA / Protein Synthesis
3. Cell transformation → increase likelihood of cancer (HPV)



4. Suppression of immune system (HIV)

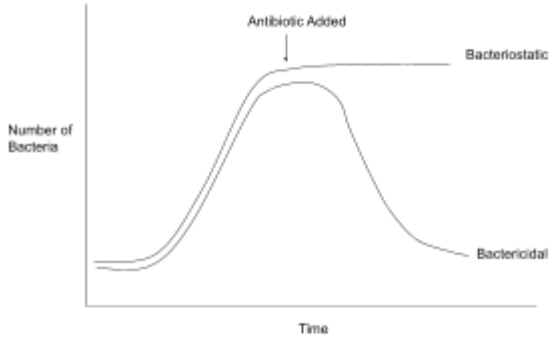
Antibiotics

- <u>Produced by fungi to kill and prevent the growth of bacteria</u>
- Target prokaryotic metabolism
- It do not interfere with host cell metabolism
- **<u>It does not act on virus</u>**

Bacteri<u>cidal</u> antibiotics: <u>kill</u> bacteria
Bacterio<u>static</u> antibiotics: <u>prevent</u> bacteria growth and replication

Broad-spectrum: affect **many** gram +ve and gram -ve <u>(Example Ampicillin and Tetracycline)</u>
Narrow-spectrum: narrower range – only +ve / -ve <u>(Example Penicillin)</u>

Antiseptics – Use on living tissue (Dettol)
Disinfectant – Use on non-living surface (Bleach)



X – Total Cell Count
✔ – Viable Cell Count

<u>Bactericidal</u> – Kill Bacteria + Bacterial number reduced

- <u>Penicillin inhibits enzyme forming amino acid linkages with peptidoglycan cell wall</u> by non-competitive inhibition
- No new cross linkage formed → weakened cell wall
- Water enter cell by osmosis
- Cell lysis
- Example: Penicillin is only effective again gram +ve → narrow spectrum

<u>Bacteriostatic</u> – No cell death only prevent replication

- <u>Affects protein synthesis</u>
- Competitive inhibitor bind to 30s of bacterial ribosome and prevent tRNA binding
- Eukaryotic cells are not affected as it is 80s instead of 70s
- Example: <u>Tetracycline</u> which is effective against gram -ve and gram +ve

Gram Positive:

Penicillin

- <u>Prevent cross link reforming</u> in <u>peptidoglycan layer</u>
- Osmosis → lysis
- *(Enzyme for reforming: transpeptidase)*

Lysosyme

- Lysosome <u>breaks down</u> exist cross link
- Osmosis → Lysis

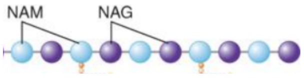
Techracyline

- Prevent protein synthesis
- By binding on ribosome
- Preventing Translation

- It does not work on Eukarotyiotic (80s ribosome vs 70s bacteria ribosome)
- It does not work on Virus (as there is no ribosome present)

Gram Negative:

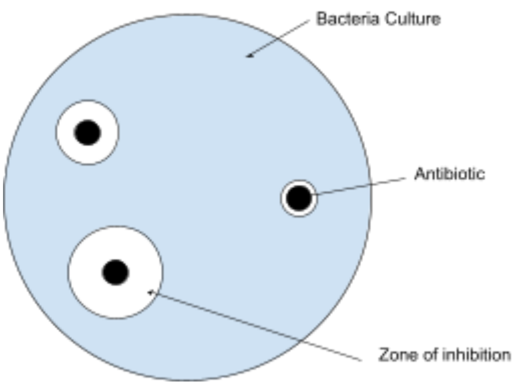
- Antibiotic cant get to peptidoglycan
- As it cannot get pass lipopolysaccharide



FYI: NAG NAM forms the peptidoglycan

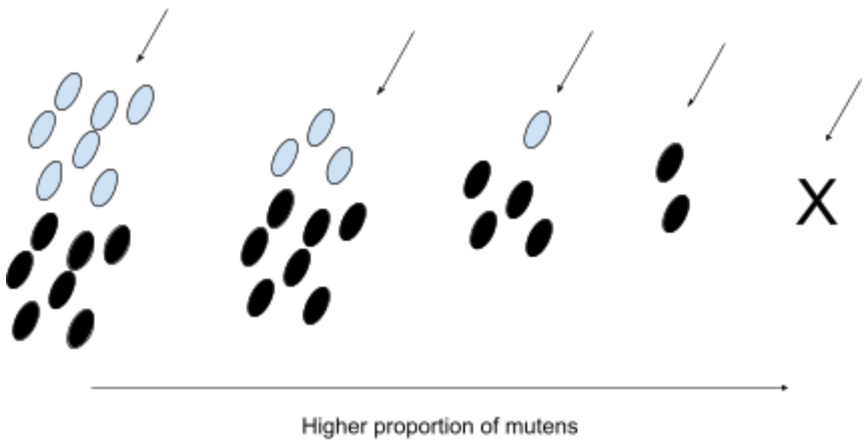
Determining bacterial sensitivity

- Spread the unknown bacterial culture on an agar plate.
- Place filter paper disks soaked in <u>different antibiotics</u> onto the agar plate.
- The antibiotics will <u>diffuse</u> outwards.
- The most effective antibiotic is determined by measuring the <u>largest clear zone</u> (the zone of inhibition).



Antibiotic Resistance

Immune – A learned response of how the body fights a specific antigen
Resistance – When a drug is no longer effective against a bacteria due to a random mutation



Resistance population have the potential to transfer their mutilated allele to other bacteria in their environment

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Immune Response

Innate Immune System

- Physical & Chemical Barrier (1st line of defence) *no cells involve*
- Phagocytosis (2nd line of defense)
- Not specific – <u>does not distinguish between pathogen</u>
- Acts immediately – no delay

Adaptive Immune System

- Specific to certain pathogen
- Involves B lymphocytes and T lymphocyte → Takes time to act

Innate Immune System (1st line of defence)

Physical and chemical barrier to pathogen

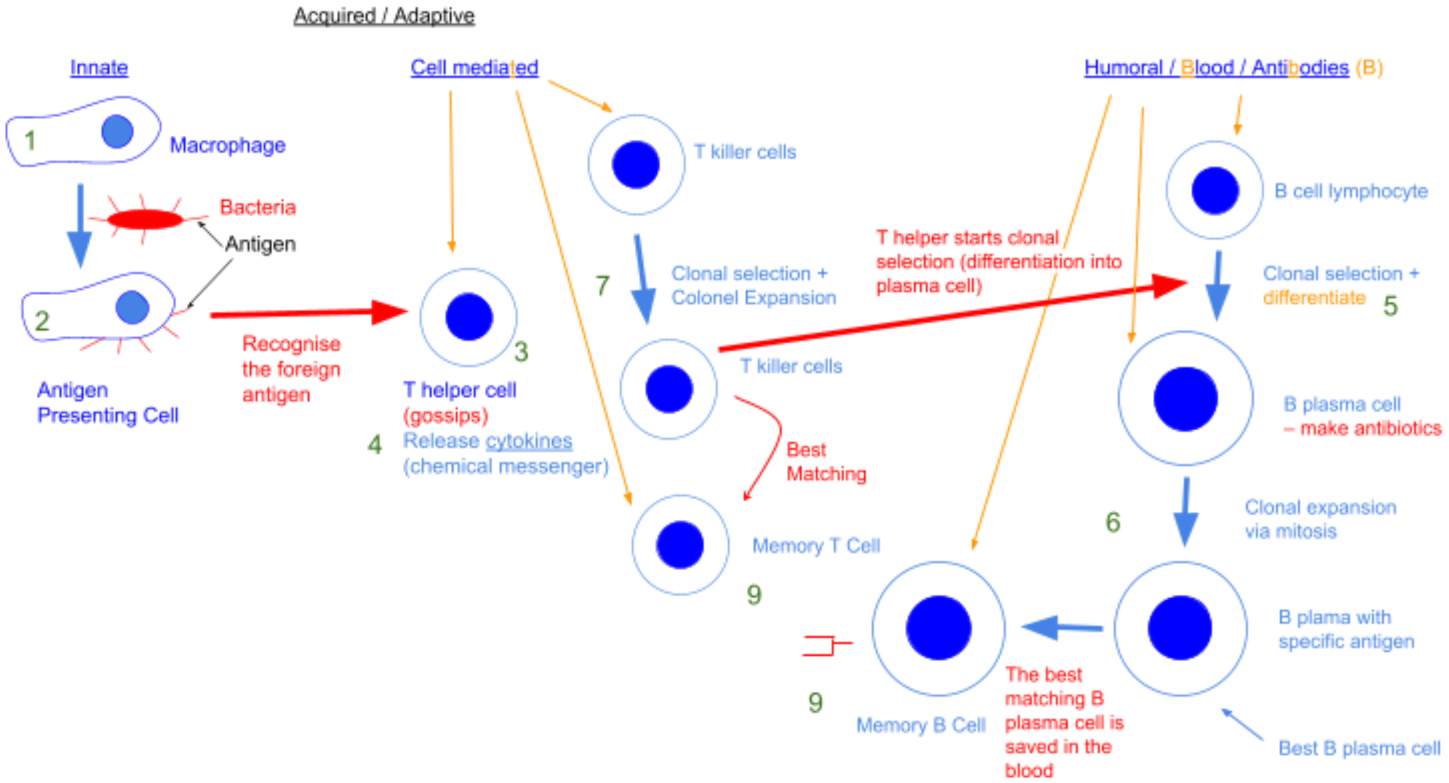
Skin	1. Collagen - Maintain by vitamin C → Strong connective tissue 2. Keratin - In epidermal cells → water impermeable 3. Skin flora - Fungi and bacteria compete interspecifically with pathogen bacteria - Not easily removed by washing - Response when touch skin is broken Response to when tough skin barrier is broken 4. Blood clotting - Steal the wound → prevent entry of microbes 5. Increase temperature - To damage site → unfavourable for microbial growth 6. <u>Inflammation</u> - Phagocyte rush to damage site - Release histamine → vasodilation → Increase blood flow → photocyte arrival - Engulf and digest pathogen - Get hot as platelet close the wound → heat flow blood cannot escape - Swelling and redness as increase blood flow
Ciliated mucous membranes	Trap microbes in inhaled air → passed up trachea → cough out / swallowed
Stomach Acid	Kill microbe ingested in food and drinks
Lysozyme	- In tears, saliva, mucus - Kill bacteria by cleaving peptidoglycan chain → weaken bacteria cell wall

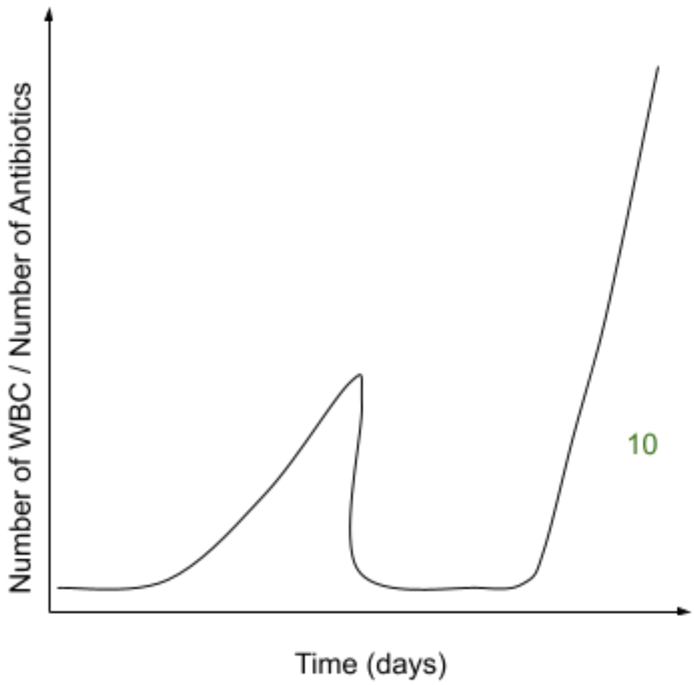
Innate Immune System – 2nd line of defence

1. Phagocyte rush to site of damage
2. Phagocyte release histamine
3. Histamine causes vasodilation which increase blood flow to the area
4. Therefore more phagocyte arrive
5. The area gets hot as platelets close the wound so heat from the blood cannot escape easily
6. Increase blood create swelling and redness

- MHC (Major Histocompatibility Complex)
- MHC Class I protein: All human Cell
- MHC Class II protein: On specialised human Cell
- Foreign antigen initiate an immune response

Adaptive Immune System – 3rd line of defense





New Bacterial Infection

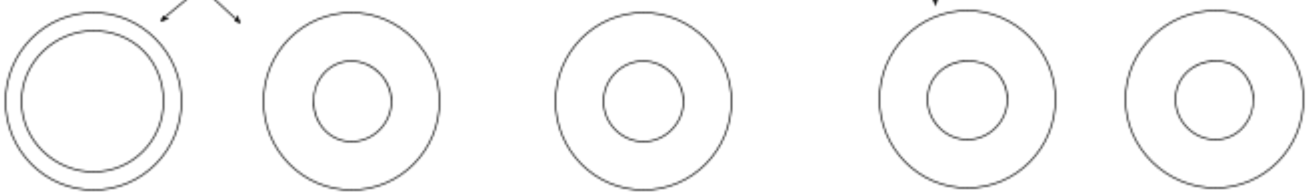
1. Macrophage in the blood release histamine and engulf + digest the bacteria and prevent the non self antigens on their surface
2. The macrophage becomes an Antigen Presenting Cell (APC)
3. The T helper recognise APC.
4. The T helper releases cytokines
    - Cause more macrophage to go go site of infectious
    - Tell B plasma cell to make antibodies
    - Cause clonal selection + expansion
5. The cytokines cause B cells to differentiate into B plasma cells
6. The plasma cells undergo clonal selection and clonal expansion by mitosis
7. The cytokines cause T killer cells to undergo clonal selection and clonal expansion by mitosis
8. The cell mediated + humoral branches of the immune system combine to fight the pathogen
9. B memory cell and T memory cell remain in low number in the blood. They activate quickly if the same pathogen infects
10. In a second infection, the memory cells undergo clonal expansion quickly

Adaptive Immune System

Humoral System
B cells
- Made and mature in Bone marrow
- Act on pathogen free in the Blood
- Produce antibodies

Cell mediaTed system
T cells (cell to cell communication)
- Mature in the Thymus
- Act on infecTed hosT cells

T Suppressor Cell (ext)
- Switch off immune system



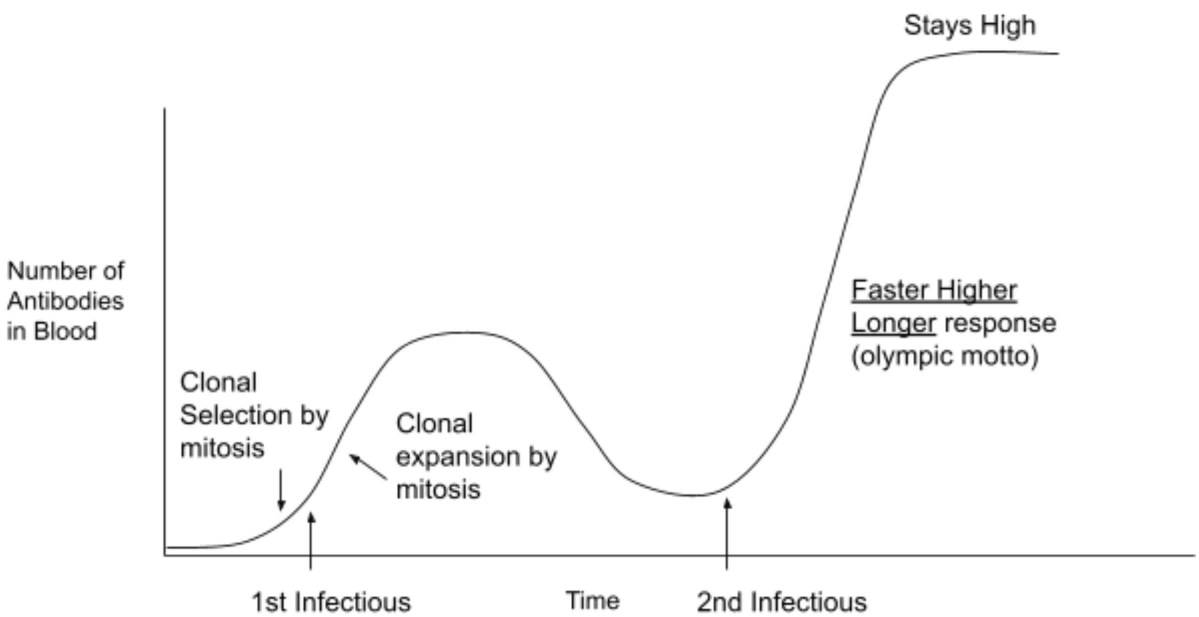
Plasma Cells
- Cytokines stimulate and activates
- Produce antibodies
- Free pathogen / antigen in blood

B Memory Cell
- To recognise + fight the same pathogen / antigen
- Faster response next time

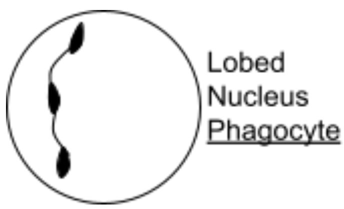
T Helper Cells
- Release Cytokines

T Killer Cells
- Kill infected host cell by causing cell lysis
- Fuse with target cell
- Release chemical to perforate cell membrane
- Coats cell chemicals, to mark for phagocytosis
- Cause lysis

T Memory Cell
- You can recognise + fight same pathogen antigen faster next time
- Dormant
- Rapid response to future infectious of same pathogen



Key Terms



Clonal selection: Finding most appropriate antibodies produced by B plasma cell against antigen by mitosis
Clonal expansion: Reproducing the best antibodies by mitosis

Antigen

Surface proteins that is recognised as non-self by the immune system and will stimulate an immune response

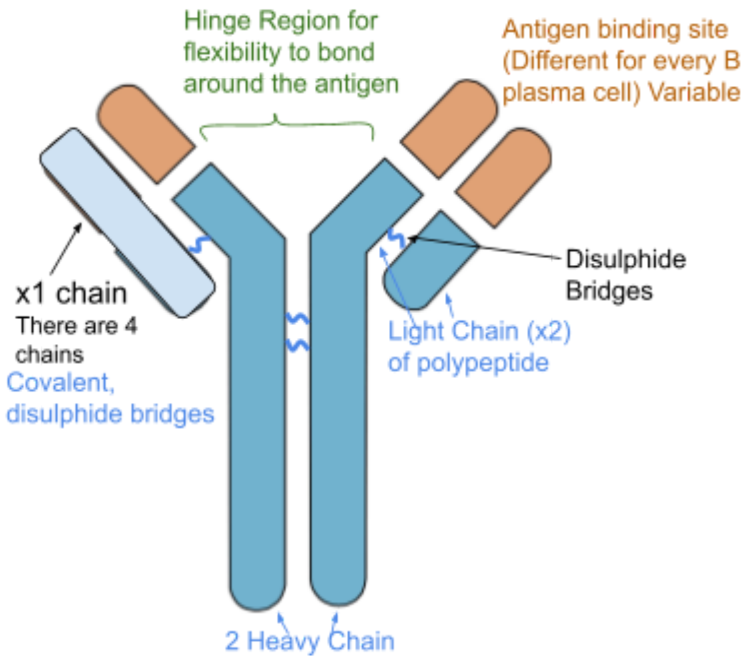
B lymphocytes

- Generate in bone marrow
- Mature in bone marrow
- Activated in spleen and lymph node
- B cell activate to plasma and memory cell

T lymphocytes

- Generated in bone marrow
- Activated in Thymus gland
- T cell activate to T killer, T helper and T memory

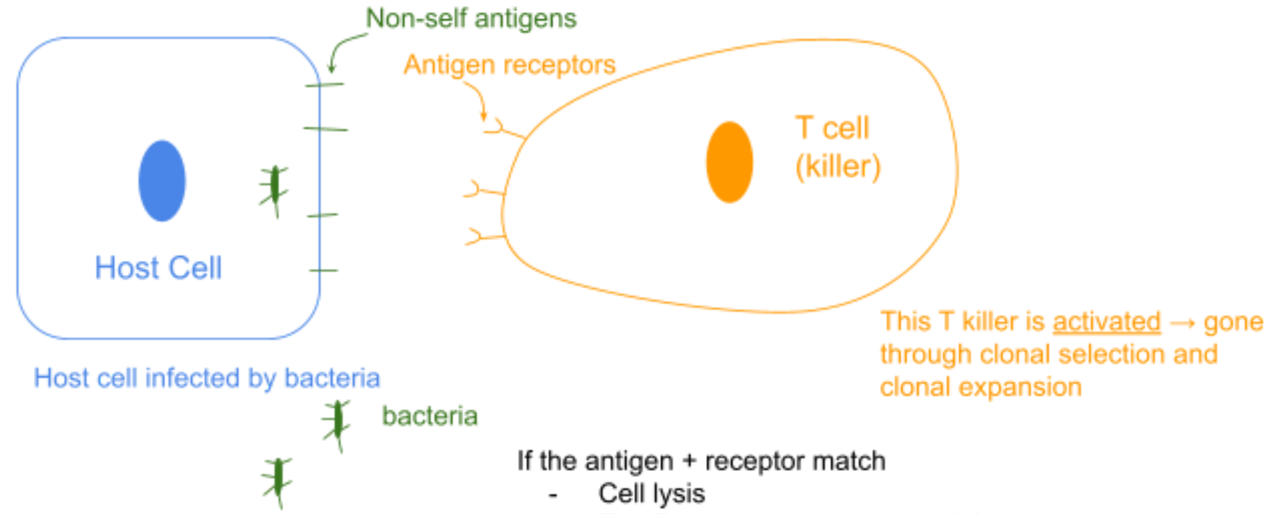
Antibodies (Y shaped Immunoglobulins [IgM/IgG])



Keywords

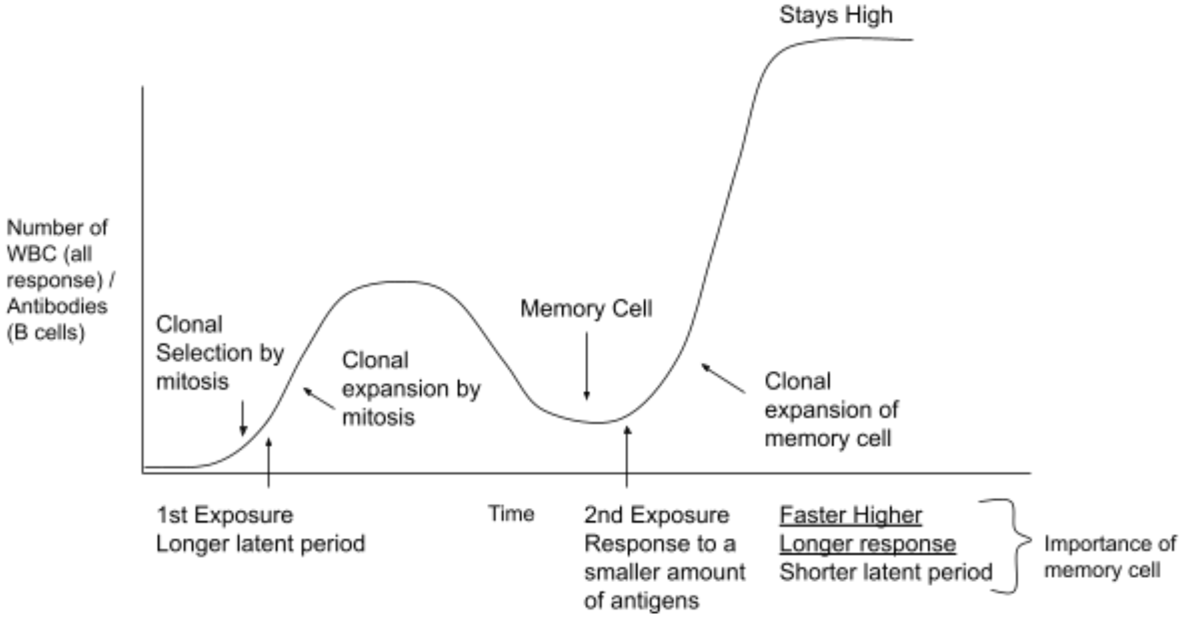
- Antibody–Antigen complex (a **<u>specific</u>** antibody bound to an antigen / complimentary antigen)
- Antibodies <u>inactive</u> (antigen cannot get into cell) antigen by:
    - <u>Agglutination</u> / clumping together to increase rate of engulfment by phagocyte
    - <u>Neutralisation</u> – prevent pathogen from binding to target cell
- B plasma cell have receptor which is complimentary to antigen (pathogen in blood)
- T killer cells complimentary to the antigen (pathogen in a host cell)

Cell Mediated (T Cells)



If the antigen + receptor match
- Cell lysis
- The T killer labels the host cell for phagocytes

- Antigen receptor on T killer cell binds to non self antigen on infected cell
- Perforin makes holes in infected cell membrane and enzyme enters
- Infected cell is destroyed



Active and Passive Immunity

Active Immunity

- Natural: natural infection (when you have it yourself)
- Induced: Have vaccination
- Individual produces antibodies by themselves
- Injections / Vaccinations

Passive Immunity

- Receives antibodies produced by another individual
- Babies fed by breast milk → Antibodies (immunoglobulin) passing through the milk
- No memory cell are made (body does not learn from it)

Vaccination

- Artificial adaptive immunity
- Expose the body to a harmless antigen (did not get the disease) for example
    - <u>Antigens isolated from the pathogen</u>
    - <u>Weakened (attenuated) strains of the pathogen</u>
    - <u>Inactive or killed pathogen</u>
    - <u>Inactivated toxin</u>

- **Low immune response → Booster Shot**
- Booster shots: For antigen <u>changing shape</u> + induce a secondary immune response for <u>greater number of memory cells</u> → so response is faster higher longer
- Emergency passive immunity: <u>Presynthesised</u> antibodies
- Low level of mutations → Low level of variation → Vaccine last longer
- <u>Immunogenic</u>: Able to generate an immune response (antigen)
- <u>Herd Immunity</u>: Enough people are immunised, the community is under protection

Human Pathogen

- 35 – 38°C
- Similar to body temperature (as it is a human pathogen)

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